Euler's Constant and Shifted Mills Constants

Arithmetic Structure, Reciprocal Series, and Proof-Transfer Limits

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5 November 2026

An integrated account of twenty ranked research items

Abstract

We integrate ten gamma-related and ten Mills-related results, reductions, and audits into one research manuscript. On the gamma side, winding versions of the canonical phase decomposition of E-function values lead to relative algebraic-closure statements for fields generated by exponential-integral values, a radical obstruction involving the Euler–Gompertz constant, and an explicit failure of relative closure under field composita. On the Mills side, shifted exponent clocks give escape from fixed polynomial relations, while hypothetical algebraicity of the least classical Mills constant imposes cubic trace identities, divisible prime gaps, and a transcendental gap exponent. A common integer-tail method proves linear independence of two reciprocal series for each shifted clock and yields a precise irrationality transfer criterion. For rational approximations to Euler's constant, we distinguish positive analytic errors from primitive arithmetic errors, prove a slow-degree denominator barrier, and record an exact check of 384 higher-degree Padé approximants. Proofs, explicit hypotheses, provenance, and reproducible integer-arithmetic certificates are kept together. The irrationality of Euler's constant and unconditional transcendence of the classical Mills constant are not established here.

Keywords. Euler–Mascheroni constant; Mills constant; E-function values; relative algebraic closure; reciprocal series; Pisot numbers; Padé approximation; exact arithmetic.

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1 Scope, notation, and the twenty-item register

This article consolidates the two ten-item rankings developed in the preceding research dialogue. The ranking labels G1–G10 and M1–M10 are retained as a navigation device. They are not twenty claims of new literature priority. The body reorganizes them by mathematical dependency and supplies proofs or precisely attributed inputs. Computational checks are distinguished from infinite statements, and conditional implications are distinguished from verification of their antecedents.

1.1 The two original targets

Euler's constant is

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

For a positive exponent sequence $C = (C_k)_{k\geq 1}$, define

$$\mathcal{W}(C) = \{A > 1 : \left\lfloor A^{C_k} \right\rfloor \text{ is prime for every } k \geq 1\}, \qquad \xi(C) = \min \mathcal{W}(C)$$

when the minimum exists. The classical least Mills constant is $M = \xi((3^k)_{k\geq 1})$. Its irrationality is an external result of Saito [4]. The conditional transcendence result under the Density Hypothesis is also external [5].

The four shifted constants considered in the dialogue are

$$A_{\pm} = \xi((3^k \pm 1)_{k\geq 1}), \qquad B_{\pm} = \xi((3^k \pm k)_{k\geq 1}).$$

Their individual transcendence is used as an input from the named working manuscripts [17, 10]; those long source proofs are not reproduced as if they were consequences of Saito's published variants theorem. The floor-sequence theorems below state their exact hypotheses on the base A, so their logical dependence is explicit. Throughout, $3^k \pm k$ means a *linear* shift, not a fixed integer offset.

1.2 Analytic notation and a necessary distinction

Write $\overline{\mathbb{Q}} = \overline{\mathbb{Q}}$, let $\mathcal{E}$ denote the ring of values of E-functions at algebraic points, and put $\mathcal{K} = \mathrm{Frac}(\mathcal{E})$. An E-function is understood in the standard arithmetic sense used in [2, 1]. The rings $\mathcal{E}$ and $\mathcal{K}$ must not be identified: the former is not a field [2]. For $a > 0$,

$$\mathrm{Ein}(a) = \int_0^a \frac{1 - e^{-t}}{t}\, dt, \qquad E_1(a) = \int_a^\infty \frac{e^{-t}}{t}\, dt,$$

and the standard connection formula [3] is

$$\mathrm{Ein}(a) = \gamma + \log a + E_1(a).$$

In particular,

$$\eta := \mathrm{Ein}(1) = \gamma + \frac{\delta}{e}, \qquad \delta := \int_0^\infty \frac{e^{-t}}{1+t}\, dt = eE_1(1). \tag{1.1}$$

1.3 Status vocabulary and provenance

Proved here means that this manuscript contains a proof from its stated standard inputs; it does not assert publication priority. *Conditional* means that the implication is established while a displayed hypothesis is not supplied. *Finite certificate* means that a specified finite computation has a reproducible exact verifier. *Imported* marks a cited theorem or a source-corpus input. The corpus documents are working manuscripts rather than independent external validation of the conclusions. Standard external results and the authors' source computations are identified separately in the bibliography. No theorem below asserts either of the two original target conclusions unconditionally.

The register on the next pages preserves every item in the user's two rankings. The item identifiers in the register link to the corresponding main results. Formula numbers and theorem labels provide the detailed dependencies.

1.4 Gamma register: G1–G10

ID	Content retained	Status and precise reach
G1	Relative algebraic closure of fields generated by positive algebraic-scale Ein values	Proved using winding and finite-phase descent. It concerns $\mathcal{K}$, not a decision of the arithmetic nature of γ.
G2	$\gamma \in \overline{\mathbb{Q}}$ forces $\delta^{1/n} \notin \mathcal{K}$ for $n \geq 2$	Proved implication. The reverse-use membership hypothesis needed to conclude transcendence of γ is not supplied.
G3	Failure of relative algebraic closure under composita	Explicit counterexample using $\overline{\mathbb{Q}}(e(\eta - q))$ for two distinct $q \in \overline{\mathbb{Q}}$.
G4	$\overline{\mathbb{Q}}(\mathrm{Ein}(a)) \cap \mathcal{E} = \overline{\mathbb{Q}}[\mathrm{Ein}(a)]$	Integral-intersection argument for every nonzero algebraic a; distinguishes the value ring from its fraction field.
G5	Algebraic independence of e, η and conditional constraints on δ	Reproved with the winding map. Conditional complexity of δ is not a contradiction to algebraicity of γ.
G6	Irrationality transfer from negative-clock reciprocal expansions	Proved for bounded integer coefficients with infinitely many nonzero terms. The requisite identity for γ is not known here.
G7	A log-free positive Padé approximation family	Infinite, unconditional rational family with strictly positive error. Primitive denominator times error remains the deciding condition.
G8	Slow-degree primitive denominator barrier	Infinite-range divergence result in its stated growth regime. It is not a theorem excluding all degrees.
G9	Exact CMF lattice and normalization costs	A specified lattice and certificate method are analyzed. Global matrix content is distinguished from the gcd of an individual output pair.
G10	384 higher-degree primitive arithmetic checks	Exact finite certificate, with code included. No extrapolation to all parameter pairs is made.

1.5 Mills register: M1–M10

ID	Content retained	Status and precise reach
M1	Two-coordinate and three-coordinate escape from fixed algebraic relations	Proved for transcendental bases. Fixed-polynomial escape is not algebraic independence of individual integer tuples.
M2	Linear independence of $1, \sum p_k^{-1}, \sum p_k^{-2}$	Proved separately for each clock. The negative clocks need no transcendence assumption; $A^2 \notin \mathbb{Q}$ suffices for the positive clocks.
M3	Continuity at the unshifted clock and normalized tail constants	A genuine limiting connection to M, with fixed-prefix stability. Transcendence does not pass through limits by itself.
M4	Divisible prime gaps and a transcendental gap exponent	Necessary consequences of hypothetical algebraicity of M. Their consistency has not been ruled out.
M5	An exact three-trace elimination polynomial	Proved on the unit-norm branch. It cannot be applied to every algebraic candidate without proving that extra restriction.
M6	Three-adic depth and prime splitting restrictions	Exact source hypotheses are retained. The conclusions do not cover arbitrary cubic Pisot numbers.
M7	A prime seed with no later divisibility return	Explicit polynomial and exact finite-state certificate of a period-85 712 orbit. This is not an all-prime infinite sequence.
M8	Quantitative conflict between prefix preservation and a shift-escape certificate	A limitation of the displayed sufficient estimates, not a universal impossibility theorem for perturbation methods.
M9	Collapse of a skipped-level corridor to the last adjacent corridor	All intermediate prime constraints are retained. A larger raw interval is not automatically an admissible lower branch.
M10	DH/RH conditional transcendence and the scope of a quasi-Riemann input	External conditional theorem and a scope comparison. No Density Hypothesis or RH is proved in this manuscript.

2 Winding embeddings and arithmetic fields

This section gives the structural results behind entries G1–G5 in the concordance. Its external analytic input is the canonical phase decomposition of E-functions and the value morphism of Fischler–Rivoal [2, Theorem 2.3 and Proposition 3.1], together with Beukers' division theorem [1]. The continuation argument and the finite-phase method were developed in the working manuscripts [13, 15, 16]; the proofs needed here are included below. These are statements about specified subfields and subrings of E-values. They neither identify the arithmetic nature of γ nor assert historical priority for their formulations.

2.1 The ambient ring and the analytic input

Throughout this section $\overline{\mathbb{Q}} = \overline{\mathbb{Q}} \subset \mathbb{C}$. Let $\mathcal{E}$ be the ring of values at algebraic points of E-functions, and put $\mathcal{K} = \mathrm{Frac}(\mathcal{E})$. One may use the usual exponential arithmetic bounds in the definition of an E-function: if $f(z) = \sum_{n\ge 0} a_n z^n/n!$, its coefficients are algebraic, their conjugates and common denominators have exponential bounds, and f satisfies a linear differential equation over $\overline{\mathbb{Q}}(z)$. All the explicit functions below belong to this class. The construction also applies to the broader Siegel convention covered in [2].

Every element of $\mathcal{E}$ can be written as $f(1)$: algebraic changes of scale and finite sums and products preserve the class of E-functions. In particular, $\overline{\mathbb{Q}} \subset \mathcal{E}$ and $e^a, e^{-a} \in \mathcal{E}$ for $a \in \overline{\mathbb{Q}}$. Define

$$\mathrm{Ein}(z) = \sum_{n\ge 1} \frac{(-1)^{n-1} z^n}{n\, n!}, \qquad \eta_a = \mathrm{Ein}(a), \qquad \eta = \eta_1. \tag{2.1}$$

For algebraic a, the function $\mathrm{Ein}(az)$ is an E-function. Its series has the required arithmetic bounds, and it satisfies $zf''' + (2+az)f'' + af' = 0$. For real $a > 0$, put

$$E_1(a) = \int_a^\infty \frac{e^{-t}}{t}\, dt, \qquad \delta = eE_1(1) = \int_0^\infty \frac{e^{-t}}{1+t}\, dt.$$

The classical connection identity gives

$$\eta_a = \gamma + \log a + E_1(a), \qquad \delta = e(\eta - \gamma). \tag{2.2}$$

No membership of γ or δ in $\mathcal{E}$ is assumed here.

We shall use the ring of finite exponential sums

$$\mathcal{R} = \left\{ \sum_{\rho \in \overline{\mathbb{Q}}} c_\rho e^{\rho X} : c_\rho \in \mathbb{C}, \quad \text{finite support} \right\}. \tag{2.3}$$

This is a domain of entire functions. Distinct exponential functions are linearly independent over $\mathbb{C}$: differentiating a finite linear relation at $X = 0$ gives a Vandermonde system.

Fix $-\pi/2 < \theta < \pi/2$. The cited phase decomposition associates to an E-function f finitely many germs f_ρ at $z = 1$ with

$$f(z) = \sum_\rho f_\rho(z) e^{\rho z}. \tag{2.4}$$

They are the canonical sums of the corresponding asymptotic components. At an exceptional direction one uses $\theta + \epsilon$ with $\epsilon > 0$ sufficiently small. For any finite collection of functions, a common such ϵ is available, and the germs are independent of sufficiently small positive changes of ϵ. The following properties of the construction are the external input: the decomposition is unique; sums and products have their corresponding componentwise and convolution decompositions; and each $e^{\rho z} f_\rho(z)$ solves an E-operator with no possible finite singularity except 0. In particular, the germs f_ρ continue along every path in $\mathbb{C}^\times$. The last property, stated just before Theorem 2.3 of [2], is essential for the argument below.

Proposition 2.1 (Continued phase embeddings)**.** *Let* M^j *mean analytic continuation along a based loop at* 1 *with counterclockwise winding number* $j \in \mathbb{Z}$ *about* 0*. For every integer* j*, the prescription*

$$\Phi_\theta^{[j]}(f(1))(X) = \sum_\rho (\mathsf{M}^j f_\rho)(1) e^{\rho X} \tag{2.5}$$

defines an injective $\overline{\mathbb{Q}}$*-algebra homomorphism* $\mathcal{E} \longrightarrow \mathcal{R}$*. It satisfies*

$$\Phi_\theta^{[j]}(\xi)(1) = \xi, \qquad \Phi_\theta^{[j]}(e^a) = e^{aX} \quad (a \in \overline{\mathbb{Q}}). \tag{2.6}$$

It extends uniquely to an embedding $\mathcal{K} \longrightarrow \mathrm{Frac}(\mathcal{R})$*.*

Proof. Suppose $f(1) = g(1)$. Beukers' division theorem supplies an E-function h such that $f - g = (z-1)h$. Choose a common sufficiently small lateral direction for f, g, h. Uniqueness of their canonical decompositions implies

$$f_\rho - g_\rho = (z-1)h_\rho$$

for each phase, with missing components interpreted as zero. Continue this identity along the same based loop. The continued h_ρ is holomorphic at the endpoint 1, since 1 is not a singularity of its differential equation. Evaluation there yields $(\mathsf{M}^j f_\rho)(1) = (\mathsf{M}^j g_\rho)(1)$. Thus (2.5) is independent of the representing E-function.

The component identities for sums give additivity. For a product $h = fg$, canonical uniqueness gives

$$h_\rho = \sum_{\alpha+\beta=\rho} f_\alpha g_\beta.$$

Continuation respects products, so this identity proves multiplicativity after evaluation at 1. Constants and pure exponentials have constant components and give the claimed normalizations. Finally, the total function f is entire and hence single-valued. Continuing (2.4) gives

$$\sum_\rho (\mathsf{M}^j f_\rho)(1) e^\rho = (\mathsf{M}^j f)(1) = f(1).$$

Evaluation at $X = 1$ is therefore a left inverse, proving injectivity. The extension to fractions is now immediate. Although evaluation is not defined on every element of $\mathrm{Frac}(\mathcal{R})$ at 1, it is defined on this embedded copy of $\mathcal{K}$: a denominator $\Phi_\theta^{[j]}(b)$, with $0 \neq b \in \mathcal{E}$, takes the nonzero value b at 1. □

Remark 2.2 (What is continued)*.* The entire function f and its value $f(1)$ do not acquire monodromy. Only its separate canonical components are continued. Nor is $\Phi_\theta^{[j]}$ defined by replacing θ with $\theta + 2\pi j$: the initial summation direction stays fixed. The continued components need not constitute a canonical asymptotic decomposition in a specified new sector. These distinctions prevent a change of analytic branch from being mistaken for a change of the number γ.

2.2 The affine winding formula for exponential integrals

Let $a > 0$ be algebraic and choose the positive real normalization. Writing $\tau = 2\pi i$, the two components of $f(z) = \mathrm{Ein}(az)$ are

$$f_0(z) = \gamma + \log a + \log z, \qquad f_{-a}(z) = e^{az} E_1(az).$$

They have phases 0 and $-a$. Their canonical identification follows either from the connection identity or from the inverse-power expansion of $e^{az}E_1(az)$; its Borel transform is $1/(a+t)$,

which has no singularity on the positive ray. Since f is entire, continuation around 0 adds $j\tau$ to f_0 and subtracts $j\tau e^{az}$ from f_{-a}. Proposition 2.1 therefore gives

$$\Phi_0^{[j]}(\eta_a) = C_{a,j} + B_{a,j}e^{-aX}, \quad \begin{cases} C_{a,j} = \gamma + \log a + j\tau, \\ B_{a,j} = e^a E_1(a) - j\tau e^a. \end{cases} \tag{2.7}$$

At $j = 1$, $B_{a,1} \neq 0$ because $e^a E_1(a) > 0$ is real. Also

$$C_{a,1} \notin \overline{\mathbb{Q}}. \tag{2.8}$$

Indeed its imaginary part is 2π. The imaginary part of an algebraic complex number is algebraic, whereas π is transcendental. This is where positivity of a supplies the reality needed for the joint theorem; no assertion about the arithmetic of $\gamma + \log a$ is used.

2.3 Finite phases and ramification

Lemma 2.3 (Finite-phase descent)**.** *Let $a_1, \dots, a_r \in \overline{\mathbb{Q}}$ be linearly independent over $\mathbb{Q}$. If $W \in \mathrm{Frac}(\mathcal{R})$ is algebraic over $\mathbb{C}(e^{a_1 X}, \dots, e^{a_r X})$, then for some integer $q \geq 1$,*

$$W \in \mathbb{C}(e^{a_1 X/q}, \dots, e^{a_r X/q}). \tag{2.9}$$

If in addition $W \in \mathcal{R}$, it lies in the Laurent polynomial ring in these variables after a possible increase of q.

Proof. Choose a rational basis extending $a_1, \dots, a_r$ for the span of all phases in a numerator and denominator representing W. Clearing their rational coordinates gives variables $X_i = e^{a_i X/q}$ and additional variables $Y_j = e^{b_j X/q}$, with

$$W \in \mathbb{C}(X_1, \dots, X_r, Y_1, \dots, Y_s).$$

These variables are algebraically independent over $\mathbb{C}$: a polynomial relation would be a linear relation among exponentials with distinct exponents. Put $F_0 = \mathbb{C}(X_1, \dots, X_r)$. The element W is algebraic over F_0, because it is algebraic over the smaller field $\mathbb{C}(X_1^q, \dots, X_r^q)$.

A purely transcendental extension $F_0(Y_1, \dots, Y_s)$ has no new elements algebraic over F_0. To check this directly, differentiate the separable minimal polynomial of such an element with respect to each Y_j. Its derivative at the element is nonzero, so each partial derivative of the element vanishes. A rational function in characteristic zero with zero derivative in a variable is independent of that variable. This proves (2.9). In the ring case, W is initially a Laurent polynomial in the X_i, Y_j; uniqueness of its Laurent coefficients removes every nonconstant Y-monomial. □

Lemma 2.4 (Preservation of degree under constants)**.** *Let $U_1, \dots, U_r$ be independent indeterminates. An irreducible polynomial in $\overline{\mathbb{Q}}(U_1, \dots, U_r)[T]$ stays irreducible over $\mathbb{C}(U_1, \dots, U_r)$. Consequently a finite extension of $\overline{\mathbb{Q}}(U_1, \dots, U_r)$ retains its degree when constants are extended to $\mathbb{C}$.*

Proof. By clearing denominators and taking a primitive part, Gauss's lemma reduces the assertion to an irreducible polynomial over the algebraically closed field $\overline{\mathbb{Q}}$. If it factored after extending constants, factors of bounded degree with suitable normalized coefficients would satisfy finitely many polynomial equations over $\overline{\mathbb{Q}}$. Nonzero normalizing coefficients can be enforced by adjoining their inverses. A solution over $\mathbb{C}$ would then give a solution over $\overline{\mathbb{Q}}$ by the weak Nullstellensatz, contradicting irreducibility. Applying this observation to the polynomial of a primitive element proves the degree statement. The same argument applies after any identification of the independent U_i with algebraically independent functions over $\mathbb{C}$. □

Lemma 2.5 (Coordinate inertia in a translated Kummer extension)**.** *Let $C_i, B_i \in \mathbb{C}$, $B_i \neq 0$, and let $Z_1, \dots, Z_r$ be independent variables with $U_i = C_i + B_i Z_i^q$. The extension*

$$\mathbb{C}(Z_1, \dots, Z_r)/\mathbb{C}(U_1, \dots, U_r)$$

is Galois with group $G = (\mathbb{Z}/q\mathbb{Z})^r$. At the generic point of $U_i = C_i$ its inertia group is precisely the ith coordinate subgroup G_i. If an intermediate extension is unramified at every such hyperplane, it is the base field.

Proof. Each coordinate transformation multiplies Z_i by an independent qth root of unity, and the degree is q^r. Along $Z_i = 0$ the function $U_i - C_i = B_i Z_i^q$ has order q. The other variables are units, and the transformations of Z_j for $j \neq i$ act nontrivially on the residue field. The transformations of Z_i act trivially on that residue field, giving inertia exactly G_i.

Let $H \subseteq G$ correspond to an intermediate field. Since G is abelian, its inertia at this divisor is the image of G_i in G/H. If all these images vanish, all G_i lie in H. They generate G, so $H = G$ and the intermediate field is the base field. □

The coordinate divisors here are valuations of abstract rational function fields. Using them does not require an exponential function to take the value zero at a finite value of X.

Theorem 2.6 (Relative algebraic closure for positive Ein families; G1)**.** *Let $a_1, \dots, a_r > 0$ be algebraic numbers linearly independent over $\mathbb{Q}$, and put*

$$F = \overline{\mathbb{Q}}(\eta_{a_1}, \dots, \eta_{a_r}).$$

The generators $\eta_{a_1}, \dots, \eta_{a_r}$ are algebraically independent, and F is relatively algebraically closed in $\mathcal{K}$:

$$\{v \in \mathcal{K} : v \text{ is algebraic over } F\} = F. \tag{2.10}$$

Proof. Use $\Phi = \Phi_0^{[1]}$ and write $C_i = C_{a_i,1}$, $B_i = B_{a_i,1}$. Its images

$$U_i = \Phi(\eta_{a_i}) = C_i + B_i e^{-a_i X}$$

are algebraically independent over $\mathbb{C}$ by the rational independence of the a_i and the exponential Vandermonde argument. A polynomial relation over $\overline{\mathbb{Q}}$ among the actual values would give one among these functions. Thus the actual values are algebraically independent over $\overline{\mathbb{Q}}$ as well.

Let $v \in \mathcal{K}$ be algebraic over F, of degree d. Its image $W = \Phi(v)$ is algebraic over $\mathbb{C}(U_1, \dots, U_r)$. Finite-phase descent supplies an integer q for which

$$W \in \mathbb{C}(Z_1, \dots, Z_r), \qquad Z_i = e^{-a_i X/q}, \qquad U_i = C_i + B_i Z_i^q.$$

The degree of $\mathbb{C}(U, W)/\mathbb{C}(U)$ is d by Lemma 2.4.

Take a primitive irreducible polynomial $P(U, T) \in \overline{\mathbb{Q}}[U_1, \dots, U_r, T]$ for v after denominators are cleared. Its leading coefficient in T times its T-discriminant is a nonzero polynomial $D(U) \in \overline{\mathbb{Q}}[U_1, \dots, U_r]$. Outside $D = 0$, the corresponding algebraic extension is unramified. For example, after inverting the leading coefficient, the polynomial is monic, and inverting the discriminant makes the algebra generated by a root finite etale. Thus every ramified divisor in the extension must lie in $D = 0$.

Suppose that the coordinate inertia at $U_i = C_i$ survived in $\mathbb{C}(U, W)$. The whole hyperplane would then be contained in $D = 0$. Expand D as a polynomial in the variables other than U_i. At least one coefficient is a nonzero polynomial in $\overline{\mathbb{Q}}[U_i]$, and its value at C_i would be zero. This would make C_i algebraic, contrary to (2.8). No coordinate inertia can survive. Lemma 2.5 implies $\mathbb{C}(U, W) = \mathbb{C}(U)$, so $d = 1$. Therefore $v \in F$. The reverse inclusion in (2.10) is immediate. □

Corollary 2.7 (No radical of a shifted Ein value)**.** *For positive algebraic a, algebraic c, and an integer $n \geq 2$, no solution of*

$$w^n = \eta_a - c$$

belongs to $\mathcal{K}$. In particular $\eta^{1/n} \notin \mathcal{K}$.

Proof. Such a w would be algebraic over $\overline{\mathbb{Q}}(\eta_a)$ and hence belong to that field by Theorem 2.6. In the rational function field $\overline{\mathbb{Q}}(U)$, the valuation of $U - c$ at $U = c$ equals 1, whereas the valuation of an nth power is divisible by n. This is impossible. □

2.4 Rational intersections and the exponential–Ein pair

Lemma 2.8 (Two affine images exclude rational denominators)**.** *Suppose $\xi \in \mathcal{E}$ has, under two injective $\overline{\mathbb{Q}}$-algebra homomorphisms $\Psi_1, \Psi_2 : \mathcal{E} \to \mathcal{R}$, images*

$$\Psi_i(\xi) = C_i + B_i e^{a_i X}, \qquad B_i a_i \neq 0, \quad C_1 \neq C_2.$$

Then ξ is transcendental and

$$\overline{\mathbb{Q}}(\xi) \cap \mathcal{E} = \overline{\mathbb{Q}}[\xi], \qquad \overline{\mathbb{Q}}[\xi] \cap \mathcal{E}^\times = \overline{\mathbb{Q}}^\times. \tag{2.11}$$

Proof. A nonconstant image rules out $\xi \in \overline{\mathbb{Q}}$, since both maps fix $\overline{\mathbb{Q}}$. Let $P, Q \in \overline{\mathbb{Q}}[T]$ be coprime and suppose $v = P(\xi)/Q(\xi) \in \mathcal{E}$. Applying Ψ_i to $Q(\xi)v = P(\xi)$ gives an identity of entire functions. If α were a root of Q distinct from C_i, some complex X_0 would satisfy $e^{a_i X_0} = (\alpha - C_i)/B_i$. Evaluation there would force $P(\alpha) = 0$, contradicting coprimality. Hence every root of Q would have to equal both C_1 and C_2. Thus Q is constant. This proves the first assertion; its reverse inclusion is immediate. If a polynomial in ξ is a unit of $\mathcal{E}$, its reciprocal is another polynomial in ξ by the first assertion. The only units of the polynomial ring $\overline{\mathbb{Q}}[\xi]$ are the nonzero constants, proving the second. □

Theorem 2.9 (The rational Ein intersection; G4)**.** *For every nonzero algebraic a,*

$$\overline{\mathbb{Q}}(\eta_a) \cap \mathcal{E} = \overline{\mathbb{Q}}[\eta_a], \qquad \overline{\mathbb{Q}}[\eta_a] \cap \mathcal{E}^\times = \overline{\mathbb{Q}}^\times. \tag{2.12}$$

In particular, η_a is transcendental and $1/\eta_a \notin \mathcal{E}$.

Proof. For positive a this follows directly from (2.7) and Lemma 2.8, using any two windings with nonzero exponential coefficient. For completeness the same argument works at complex algebraic a, with the following branch bookkeeping. Choose one admissible canonical direction for $\mathrm{Ein}(az)$. Its two phases are $0, -a$, and its components have the form

$$f_0(z) = C + \log z, \qquad f_{-a}(z) = e^{az}\big(\mathrm{Ein}(az) - C - \log z\big),$$

where C is fixed by the chosen canonical sum and the compatible logarithm. This is the connection identity analytically continued from a nonsingular summation ray. Equivalently, the inverse-power component has Borel transform $1/(a + t)$; if a ray meets its pole, one fixes either adjacent nonsingular ray and the compatible branch. Only the existence of this constant C, not its arithmetic nature, is needed here.

Write $B = e^a(\eta_a - C)$. Since $\mathrm{Ein}(az)$ and e^{az} are single-valued, continuing the components gives

$$\Phi_\theta^{[j]}(\eta_a) = C + j\tau + (B - j\tau e^a)e^{-aX}.$$

The coefficient $B - j\tau e^a$ vanishes for at most one integer j. Choose two distinct windings avoiding it. The omitted values $C + j\tau$ differ, so Lemma 2.8 applies. □

The assertion $1/\eta_a \notin \mathcal{E}$ must not be upgraded to $1/\eta_a \notin \mathcal{K}$. In fact $1/\eta_a \in \mathcal{K}$ by the definition of a fraction field. The radical exclusion in Corollary 2.7, in contrast, is genuinely a statement about the larger field $\mathcal{K}$.

Theorem 2.10 (The exponential–Ein pair; G5)**.** *The numbers e and η are algebraically independent over $\overline{\mathbb{Q}}$. If $\gamma \in \overline{\mathbb{Q}}$, then δ is transcendental over $\overline{\mathbb{Q}}(\eta)$.*

Proof. Put $C = \gamma + \tau$ and $B = \delta - \tau e$. Under $\Phi_0^{[1]}$,

$$e \longmapsto e^X, \qquad \eta \longmapsto C + Be^{-X}.$$

Here $C \notin \overline{\mathbb{Q}}$ by (2.8). Suppose that $\sum_{r=0}^{d} e^r P_r(\eta) = 0$, with $P_r \in \overline{\mathbb{Q}}[T]$ and $P_d \neq 0$. Apply the embedding and divide by e^{dX}. On letting real $X \to +\infty$, all terms with $r < d$ tend to zero, while the remaining term tends to $P_d(C)$. This is nonzero because C is transcendental. The contradiction proves algebraic independence.

If $\gamma = q \in \overline{\mathbb{Q}}$, then $\delta = e(\eta - q)$ and $\eta - q \neq 0$. Algebraicity of δ over $\overline{\mathbb{Q}}(\eta)$ would imply algebraicity of $e = \delta/(\eta - q)$ over that field, contradicting the first assertion. □

The independence of this pair is an application of the established E-function framework, not an assertion that the general Siegel–Shidlovskii phenomenon is new here. Its role is to prevent the conditional identity $\delta = e(\eta - q)$ from being mistaken for a contradiction: if $\gamma = q$ were algebraic, the required transcendence of δ over $\overline{\mathbb{Q}}(\eta)$ would be mathematically consistent with the preceding statements.

2.5 A radical criterion for Euler's constant

Theorem 2.11 (The Euler–Gompertz radical obstruction; G2)**.** *For every integer $n \geq 2$,*

$$\gamma \in \overline{\mathbb{Q}} \quad \Longrightarrow \quad \delta^{1/n} \notin \mathcal{K}. \tag{2.13}$$

Consequently, an independent proof that $\delta^{1/n} \in \mathcal{K}$ for some $n \geq 2$ would prove that γ is transcendental.

Proof. Assume $\gamma = q \in \overline{\mathbb{Q}}$ and that a root v of $v^n = \delta$ belongs to $\mathcal{K}$. The number $e^{1/n}$ is an E-value, so $w = v/e^{1/n} \in \mathcal{K}$. Equation (2.2) gives

$$w^n = \eta - q,$$

contrary to Corollary 2.7. Different choices of the root differ by an algebraic root of unity, so the conclusion is independent of that choice. The final assertion is the contrapositive. □

Remark 2.12 (The missing arithmetic membership). No inclusion $\delta^{1/n} \in \mathcal{K}$ is proved in this paper. In particular, the theorem is not an irrationality proof for γ. The same argument shows, for every $q \in \overline{\mathbb{Q}}$, that

$$\big(e(\eta - q)\big)^{1/n} \notin \mathcal{K} \qquad (n \geq 2).$$

Thus the exclusion alone does not distinguish the actual Euler constant from a hypothetical algebraic parameter in the identity $e(\eta - q)$. To use the criterion, the additional membership must come from an independent property of the actual constant δ.

2.6 A compositum counterexample

We first record a one-variable variant of the finite-cover argument. Unlike Theorem 2.6, it does not require either omitted value to be transcendental.

Lemma 2.13 (Two affine images exclude finite algebraic extensions)**.** *Let $\xi \in \mathcal{E}$. Suppose two embeddings of the form in Proposition 2.1 give*

$$\Psi_i(\xi) = C_i + B_i e^{a_i X}, \qquad a_i \in \overline{\mathbb{Q}}^{\times}, \quad B_i \neq 0, \quad C_1 \neq C_2.$$

Then $\overline{\mathbb{Q}}(\xi)$ is relatively algebraically closed in $\mathcal{K}$.

Proof. The nonconstant images imply that ξ is transcendental. Let $L = \overline{\mathbb{Q}}(\xi, v)$ for an element $v \in \mathcal{K}$ algebraic of degree d over $\overline{\mathbb{Q}}(\xi)$. Identify the transcendental generator ξ with an indeterminate U. After extending constants to $\mathbb{C}$, this gives a function field $L_{\mathbb{C}}$ of degree d over $\mathbb{C}(U)$ by Lemma 2.4.

For either embedding, finite-phase descent places the image of v in $\mathbb{C}(Z_i)$ with $Z_i = e^{a_i X/q_i}$ and $U = C_i + B_i Z_i^{q_i}$ for some $q_i \geq 1$. The comparison is made over the same abstract coordinate U; it does not identify the two functions $\Psi_i(v)$ inside their ambient exponential fields.

If $d > 1$, the first realization must be ramified at $U = C_1$. Indeed, in the one-coordinate case of Lemma 2.5, the inertia group at that point is the entire cyclic group, and its image in every nontrivial intermediate extension is nontrivial. On the other hand, the second realization is unramified at every finite point other than C_2. One may see this directly by solving $Z_2^{q_2} = (U - C_2)/B_2$ locally at a nonzero right-hand side, where its roots are simple. Since $C_1 \neq C_2$, the same abstract extension would be both ramified and unramified at C_1, a contradiction. Therefore $d = 1$ and $v \in \overline{\mathbb{Q}}(\xi)$. □

Theorem 2.14 (Relative closure is not preserved by composita; G3)**.** *For every $q \in \overline{\mathbb{Q}}$ put*

$$d_q = e(\eta - q), \qquad F_q = \overline{\mathbb{Q}}(d_q).$$

Each F_q is relatively algebraically closed in $\mathcal{K}$. If $q \neq r$, however, then

$$F_q F_r = \overline{\mathbb{Q}}(e, \eta), \tag{2.14}$$

and this compositum is not relatively algebraically closed in $\mathcal{K}$. More precisely, $e^{1/n} \in \mathcal{K} \setminus \overline{\mathbb{Q}}(e, \eta)$ for every $n \geq 2$.

Proof. All d_q belong to $\mathcal{E}$. Formula (2.7) at $a = 1$ gives, without assuming $\gamma \in \mathcal{E}$,

$$\Phi_0^{[j]}(d_q) = (\gamma - q + j\tau)e^X + \delta - j\tau e. \tag{2.15}$$

In this formula γ and δ are complex coefficients of the image, not additional presumed elements of the domain of Φ. The coefficient of e^X vanishes for at most one integer j. Choose two distinct other windings. Their constant terms $\delta - j\tau e$ differ. Lemma 2.13 proves relative algebraic closure of F_q.

For distinct q, r, the identities

$$e = \frac{d_q - d_r}{r - q}, \qquad \eta = q + \frac{d_q}{e}$$

prove (2.14). The reverse containment follows from the definitions of d_q, d_r. By Theorem 2.10, $\overline{\mathbb{Q}}(e, \eta)$ is a rational function field in the two independent generators e, η. The polynomial $T^n - e$ is irreducible over this field: view it as $T^n - U$ over $\overline{\mathbb{Q}}(V)(U)$ and apply Eisenstein's criterion at the prime U in $\overline{\mathbb{Q}}(V)[U]$. Thus $e^{1/n} \notin \overline{\mathbb{Q}}(e, \eta)$, although $e^{1/n} \in \mathcal{E} \subset \mathcal{K}$. This proves the failure of relative algebraic closure of the compositum. □

The same affine images, by Lemma 2.8, also give $\overline{\mathbb{Q}}(d_q) \cap \mathcal{E} = \overline{\mathbb{Q}}[d_q]$ and $\overline{\mathbb{Q}}[d_q] \cap \mathcal{E}^\times = \overline{\mathbb{Q}}^\times$. If $\gamma = q$ were algebraic, then $\delta = d_q$ would be a transcendental nonunit of $\mathcal{E}$, consistently with these statements. The compositum example is a concrete reason that separate closure theorems cannot be combined without a new proof of closure for their jointly generated field.

Remark 2.15 (Differences of phase images are not new E-values). For $j \neq 0$,

$$\Phi_0^{[j]}(\eta) - \Phi_0^{[0]}(\eta) = j\tau(1 - ee^{-X}) \neq 0.$$

This difference vanishes at $X = 1$. It cannot lie in the image of any continued phase embedding of an E-value: evaluation would force that value to be 0, whose image is identically zero. Subtracting two different embeddings therefore does not provide an inverse-image construction for π, γ, or δ. All membership assertions used in this section were established in the original ring or field before applying an embedding.

3 Shifted prime clocks and fixed polynomial relations

For a positive real sequence $C = (C_k)_{k\geq 1}$, define

$$\mathcal{W}(C) = \{A > 1 : \left\lfloor A^{C_k} \right\rfloor \text{ is prime for every } k \geq 1\}.$$

When this set is nonempty, its least element is denoted by $\xi(C)$. The starting index is part of this definition. In particular,

$$M = \xi((3^k)_{k\geq 1}), \qquad A_\pm = \xi((3^k \pm 1)_{k\geq 1}), \qquad B_\pm = \xi((3^k \pm k)_{k\geq 1}).$$

Existence for the clocks used below follows from Saito's existence theorem, since their successive exponent ratios are at least 2 [5, Theorem 1.3]. The irrationality of the original M is an external result [4]; its unconditional transcendence is not proved here.

Status of the named shifted examples. The transcendence statements for $A_\pm$ and $B_\pm$ were inputs to the research discussion. Their source proofs are in the internal manuscripts [17, 10], with additional extensions and audits in [18]. We do not reproduce or independently certify those complete proofs in this paper. The theorems below are therefore formulated for an arbitrary *transcendental* real base $A > 1$; their proofs are independent of the status of those four particular examples. Primality is unnecessary for the polynomial-avoidance assertions themselves.

3.1 Two coordinates for a constant shift

Theorem 3.1 (Fixed polynomial avoidance for constant shifts)**.** *Let $A > 1$ be transcendental, let σ be a nonzero integer, and assume $3^k + \sigma > 0$ at the indices under consideration. Put $p_k = \left\lfloor A^{3^k+\sigma} \right\rfloor$. For every fixed $h \geq 1$,*

$$\frac{p_{k+h}}{p_k^{3^h}} \longrightarrow A^{\sigma(1-3^h)}.$$

For every nonzero $P \in \overline{\mathbb{Q}}[X, Y]$,

$$P(p_k, p_{k+h}) \neq 0 \qquad \textit{for all sufficiently large } k.$$

The threshold may depend on A, σ, h, P.

Proof. Write $d = 3^h$ and $L = A^{\sigma(1-d)}$. The elementary estimate

$$p_k = A^{3^k+\sigma}(1 + O(A^{-3^k-\sigma}))$$

proves the ratio limit. Since $\sigma(1-d)$ is a nonzero integer, L is transcendental. Write $P(X,Y) = \sum c_{ij}X^iY^j$ and let $D = \max\{i + dj : c_{ij} \neq 0\}$. Then

$$\frac{P(p_k, p_{k+h})}{p_k^D} \longrightarrow \sum_{i+dj=D} c_{ij}L^j.$$

For a given j, the equation $i + dj = D$ specifies at most one i. The polynomial on the right is consequently nonzero, and cannot vanish at the transcendental number L. □

For the two constant shifts in the comparison register, this gives

$$\frac{p_{+,k+1}}{p_{+,k}^3} \longrightarrow A_+^{-2}, \qquad \frac{p_{-,k+1}}{p_{-,k}^3} \longrightarrow A_-^2.$$

The result excludes a *fixed* algebraic-coefficient recurrence on infinitely many pairs. It does not say that an individual pair of integers is algebraically independent.

3.2 Three coordinates for a linear shift

Theorem 3.2 (Fixed polynomial avoidance for linear shifts)**.** *Let $A > 1$ be transcendental and $\varepsilon \in \{1, -1\}$. Put $q_k = \left\lfloor A^{3^k+\varepsilon k} \right\rfloor$. Then*

$$\frac{q_{k+2}q_k^3}{q_{k+1}^4} \longrightarrow A^{-2\varepsilon}.$$

For every nonzero $P \in \overline{\mathbb{Q}}[X, Y, Z]$,

$$P(q_k, q_{k+1}, q_{k+2}) \neq 0 \qquad \textit{for all sufficiently large } k.$$

Proof. The exponent in the ratio is exactly

$$(3^{k+2} + \varepsilon(k+2)) + 3(3^k + \varepsilon k) - 4(3^{k+1} + \varepsilon(k+1)) = -2\varepsilon,$$

which proves the first claim.

For a monomial $X^iY^jZ^\ell$, define

$$W = i + 3j + 9\ell, \qquad V = i + j + \ell, \qquad U = j + 2\ell.$$

Its unfloored value is $A^{W3^k+\varepsilon Vk+\varepsilon U}$. The linear map $(i, j, \ell) \mapsto (W, V, U)$ is injective because

$$\det \begin{pmatrix} 1 & 3 & 9 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} = 4.$$

Among the finitely many monomials of P, first maximize W, then maximize εV. In this leading group, distinct monomials have distinct U, so their remaining coefficient is a nonzero Laurent polynomial in A, with algebraic coefficients. It is nonzero by transcendence of A.

Terms with smaller W are exponentially smaller on the 3^k scale; terms with the same W but smaller εV are exponentially smaller on the k scale. Replacing unfloored powers by floors has relative error $O(A^{-3^k+O(k)})$ in every fixed monomial. After normalization by the leading $A^{W3^k+\varepsilon Vk}$, these errors also tend to zero. The normalized polynomial therefore tends to the nonzero coefficient just identified. □

The extra coordinate is explained by the three scales $3^k, k, 1$. These theorems apply separately to each chosen base. They do not assert joint algebraic independence of A_+, A_-, B_+, B_-.

3.3 Continuity at the original clock

The following continuity theorem is an analytic connection with the actual minimum; it is not an arithmetic limit theorem. The argument, recorded in [18], is included because finite initial constraints cannot be discarded in this problem.

Lemma 3.3 (Uniform continuation from a large prime)**.** *There is P_0 such that every prime $p \geq P_0$ at any level of a clock whose successive ratios lie in $[5/2, 4]$ has an infinite prime continuation with strictly nested closed inverse-power cells.*

Proof. The Baker–Harman–Pintz bound supplies a constant K such that $\mathrm{nextprime}(x) - x \leq Kx^{21/40}$ for all sufficiently large x [6]. Uniformly for $5/2 \leq c \leq 4$,

$$(p+1)^c - p^c \geq cp^{c-1}, \qquad \frac{cp^{c-1}}{p^{21c/40}} = cp^{19c/40-1} \geq \frac{5}{2}p^{3/16}.$$

Hence, for all sufficiently large p, a prime q can be chosen with $p^c < q$ and $q + 1 < (p+1)^c$. Iteration gives strictly nested closed root cells, whose lengths tend to zero. Their common point lies below each upper endpoint, because the next cell already does, and thus has the prescribed prime floors. □

Theorem 3.4 (Continuity and normalized tails)**.** *For $-1 \leq t \leq 1$, put*

$$F(t) = \xi((3^k - t)_{k\geq 1}).$$

Then $F(t) \to M$ as $t \to 0$. Consequently,

$$\xi((3^{k+h} - 1)_{k\geq 1})^{3^h} \longrightarrow M, \qquad \xi((3^{k+h} + 1)_{k\geq 1})^{3^h} \longrightarrow M.$$

At each fixed finite set of indices, the prime outputs of these normalized minima on their own perturbed clocks eventually coincide with the outputs of M.

Proof. The ratios $(3^{k+1} - t)/(3^k - t)$ lie in $[5/2, 4]$. Lemma 3.3, starting at one fixed sufficiently large first prime, gives a uniform upper bound on $F(t)$. The first floor gives $F(t) \geq 2^{1/4} > 1$.

Suppose $t_n \to 0$ and $F(t_n) \to A$. If $A^{3^j} = m$ were an integer, $m \geq 2$, then

$$F(t_n)^{3^{j+2}-t_n} \longrightarrow m^9.$$

Its prime floor would eventually be either $m^9 - 1$ or m^9, both composite. Thus every A^{3^j} is nonintegral. Continuity of the floor at those points shows that $A \in \mathcal{W}(3^k)$, so $A \geq M$. This proves $\liminf F(t) \geq M$.

Write $p_j = \left\lfloor M^{3^j} \right\rfloor$. No M^{3^j} is integral: otherwise the next prescribed power is composite. The closed prime cells are strictly nested, since

$$p_j^3 < p_{j+1}, \qquad p_{j+1} + 1 < (p_j + 1)^3.$$

Equality at the second endpoint would give the composite integer $(p_j + 1)^3 - 1$. Given $\epsilon > 0$, choose n with $p_n \geq P_0$ and $(p_n + 1)^{1/3^n} < M + \epsilon$. Its closed cell lies strictly inside all earlier prime cells. This finite collection of strict containments persists after replacing 3^j by $3^j - t$, for small enough $|t|$. Continue from the same p_n using Lemma 3.3. The resulting admissible base gives

$$F(t) \leq (p_n + 1)^{1/(3^n - t)}.$$

Taking upper limits, then $\epsilon \downarrow 0$, proves continuity.

The increasing change of variable $A \mapsto A^d$ gives the exact scaling identity $\xi(dC) = \xi(C)^{1/d}$ for $d > 0$. Thus the two normalized tails are $F(3^{-h})$ and $F(-3^{-h})$. Finally, $F(t)^{3^j - t} \to M^{3^j}$, and the nonintegrality already proved gives stability of any fixed finite set of floors. □

The unnormalized tail minima tend to 1. Neither this convergence nor a common finite prime prefix transfers transcendence to M.

4 Arithmetic of a hypothetically algebraic Mills tower

4.1 The external reduction and the least-prime property

Theorem 4.1 (Saito's reduction: external input)**.** *If M is algebraic, then for some integer $m \geq 1$, $\beta = M^{3^m}$ is a totally real cubic Pisot number. Its conjugates can be written*

$$\beta,\ -\rho,\ \tau, \qquad 0 < \sigma := |\tau| < \rho < 1, \qquad \rho \leq \min(\sigma^{17/23}, \beta^{-17/40}).$$

This is the specialization of [5, Theorem 1.7] used throughout this section. All statements below about the actual Mills tower are implications of $M \in \overline{\mathbb{Q}}$, unless explicitly stated otherwise. Put

$$p_r = \left\lfloor \beta^{3^r} \right\rfloor, \quad g_r = p_{r+1} - p_r^3, \quad x_r = p_r^3, \quad C = \mathrm{N}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta), \quad N = |C|.$$

On a sufficiently late tail,

$$p_r = \mathrm{Tr}(\beta^{3^r}),$$

because $-\rho^{3^r} + \tau^{3^r} \in (-1, 0)$.

Lemma 4.2 (Eventual least-prime choice)**.** *For the actual least Mills constant, without assuming algebraicity, each sufficiently late prime is the least prime strictly above the cube of its predecessor. In the notation above,*

$$p_{r+1} = \mathrm{nextprime}(p_r^3).$$

Proof. Every sufficiently large prime has an infinite strict cubic continuation by Lemma 3.3. If a prime q satisfies $p_r^3 < q < p_{r+1}$, its inverse-power cell is contained in that of p_r, and all earlier classical cells are preserved. Continue from q. The upper endpoint of the replaced cell is at most $p_{r+1}^{1/3^{m+r+1}} < M$. The continued base is therefore smaller than M and satisfies every original prime-floor condition, contradicting minimality. □

4.2 Divisibility and the exact exponent

The following arithmetic consequences were recorded in [11]. Their elementary proofs are supplied here.

Theorem 4.3 (Gap divisibility and transcendental exponent)**.** *Under $M \in \overline{\mathbb{Q}}$, the sufficiently late gap sequence satisfies*

$$g_r \mid g_{r+1}, \qquad \frac{9g_{r+1}}{g_r^3} \longrightarrow 1, \qquad \frac{g_r}{p_r} \longrightarrow \infty.$$

Moreover,

$$g_r \sim 3x_r^{\kappa}, \qquad \kappa = \frac{2 + \log\rho/\log\beta}{3},$$

where κ is transcendental and

$$\frac{1}{2} + \frac{\log N}{6\log\beta} < \kappa < \frac{21}{40}.$$

Proof. At a given index, let X_1, X_2, X_3 be the 3^r-th powers of the three conjugates, and write

$$p = X_1 + X_2 + X_3, \quad b = X_1X_2 + X_1X_3 + X_2X_3, \quad c = X_1X_2X_3.$$

These are integers. Cubing the roots gives the Newton state map

$$p' = p^3 - 3bp + 3c, \qquad b' = b^3 - 3cbp + 3c^2, \qquad c' = c^3. \tag{4.1}$$

If $h = c - bp = g_r/3$, then

$$h = -(X_1 + X_2)(X_1 + X_3)(X_2 + X_3).$$

Thus

$$h' = h\,\mathcal{F}, \qquad \mathcal{F} = \prod_{i<j}(X_i^2 - X_iX_j + X_j^2).$$

The multiplier is a positive integer: it is a symmetric polynomial in algebraic integers, and all three roots are real. Hence $g_r \mid g_{r+1}$.

Since $|\tau| < \rho < 1 < \beta$, the same product identity yields

$$p_r \sim \beta^{3^r}, \qquad h_r \sim (\beta^2\rho)^{3^r}, \qquad g_r \sim 3(\beta^2\rho)^{3^r}.$$

This proves the cubic gap asymptotic and the formula for κ. The absolute norm satisfies

$$1 \le N = \beta\rho\sigma < \beta\rho^2.$$

Consequently $\rho > \sqrt{N/\beta}$, which gives the stated strict lower bound for κ, and also $\beta\rho > 1$. The latter implies $g_r/p_r \to \infty$. Saito's bound initially gives $\kappa \le 21/40$.

It remains to determine the arithmetic nature of the exponent. If $t = \log\rho/\log\beta$ were rational, it would be negative, so $\beta^u\rho^v = 1$ for some positive integers u, v. Equivalently, $\beta^u(-\rho)^v = (-1)^v$. The Galois group of an irreducible cubic contains a three-cycle; choose the orientation sending β to $-\rho$ and $-\rho$ to τ. Applying it and taking absolute values gives $\rho^u\sigma^v = 1$, impossible since both factors are below 1. Thus t is irrational. If t were algebraic, the Gelfond–Schneider theorem would make $\beta^t = \rho$ transcendental, again a contradiction [7]. Therefore t, and hence κ, is transcendental. The rational endpoint $21/40$ is excluded. □

In particular, $g_r \mid g_{r+s}$ and

$$p_{r+s} \equiv p_r^{3^s} \pmod{g_r} \qquad (s \ge 1).$$

These necessary conditions are not mutually contradictory as presently understood. The theorem is not an unconditional transcendence proof for M.

4.3 A fixed three-term identity in the unit branches

Proposition 4.4 (Eliminating the second symmetric coordinate)**.** *Let p, q, r be the traces of the first, third and ninth powers of three algebraic numbers whose elementary symmetric functions are integers, and whose product is c. Then*

$$\begin{aligned} P_c(p,q,r) := {} & q(p^3 + 3c - q)^3 + 9p^3(r - q^3 - 3c^3) \\ & - 27cp^3q(p^3 - q) = 0. \end{aligned} \tag{4.2}$$

If the starting norm is $c = \pm 1$, the same polynomial applies at every consecutive triple of sparse traces.

Proof. Let b be the initial second symmetric coordinate. Equation (4.1) gives

$$q = p^3 - 3bp + 3c, \quad b' = b^3 - 3cbp + 3c^2, \quad r = q^3 - 3b'q + 3c^3.$$

Substitution into (4.2) cancels all terms. Odd cubing preserves $c = \pm 1$, proving the final statement. □

For a nonunit norm, the corresponding polynomial is $P_{C^{3^r}}$; its coefficients vary with r. Neither the unit assumption nor a fixed polynomial can therefore be imposed on all hypothetical algebraic Mills candidates. Even in the unit branches, transferring this identity to one of the shifted sequences requires a further argument.

5 Local restrictions and an exact nonreturn certificate

The next two results retain the hypotheses of the source [14]. Their scope is a specified unit branch, not all possible algebraic Mills constants.

5.1 Linear 3-adic depth

Theorem 5.1 (Exact depth at a unit fixed state)**.** *Let $\varepsilon \in \{1, -1\}$ and let*

$$f(X) = X^3 - aX^2 + bX + \varepsilon$$

be irreducible, with $a \equiv \varepsilon \pmod 3$ and $b \equiv -1 \pmod 3$. Let T_r, B_r be the first and second elementary symmetric functions of the 3^r-th powers of its roots. If $s = v_3(a + \varepsilon b)$, then s is a finite positive integer and

$$v_3(T_r - \varepsilon) = r + s \qquad (r \geq s).$$

Proof. The norm is $-\varepsilon$ and is unchanged by odd powering. Put

$$x_r = T_r - \varepsilon, \quad y_r = B_r + 1, \quad z_r = x_r + \varepsilon y_r, \quad w_r = x_r - \varepsilon y_r.$$

Expansion of (4.1) gives

$$z_{r+1} = z_r(3 + 3\varepsilon w_r + x_r^2 - \varepsilon x_r y_r + y_r^2),$$
$$w_{r+1} = w_r(9 + 3\varepsilon w_r + x_r^2 + \varepsilon x_r y_r + y_r^2).$$

Both x_r, y_r are divisible by 3 at every stage. The first multiplier is 3 (mod 9); the second is divisible by 9. Since $z_0 = a + \varepsilon b \neq 0$, we get

$$v_3(z_r) = r + s, \qquad v_3(w_r) \geq 2r + 1.$$

Nonvanishing of z_0 follows from $f(-\varepsilon) = -a - \varepsilon b$ and irreducibility. For $r \geq s$, the two valuations are different, and $2x_r = z_r + w_r$ proves the assertion. □

5.2 A splitting restriction

Lemma 5.2 (Projective return)**.** *Suppose $p \nmid c\,\mathrm{disc}(f)$, where c is the norm of the roots, and $p \mid T_r$. In a finite splitting field modulo p, let Y_i be the roots raised to 3^r. If the multiplicative subgroup generated by the ratios Y_i/Y_j has order prime to 3, then $p \mid T_{r+jL}$ for all $j \geq 0$, for some $L \geq 1$.*

Proof. Choose L with $3^L \equiv 1$ modulo the exponent of the ratio subgroup. At each L-step, the three powered roots are multiplied by one common nonzero scalar. Their zero sum therefore recurs. □

Theorem 5.3 (Complete splitting in a specified norm--1 branch)**.** *Let β be a totally real cubic Pisot number whose small conjugate of largest modulus is negative. Suppose its polynomial is*

$$f(X) = X^3 - aX^2 + bX + 1, \qquad a \equiv 1,\ b \equiv -1 \pmod 3, \qquad v_3(a+b) = 1.$$

If $\lfloor \beta^{3^r} \rfloor$ is eventually prime, then f splits completely modulo every sufficiently late such prime.

Proof. Eventually the floor is T_r, and it exceeds every fixed prime divisor of $3\,\mathrm{disc}(f)$. For a prime $p = T_r$, Theorem 5.1 gives $p \equiv 1 \pmod 3$ and $v_3(p-1) = r+1$.

If f is irreducible modulo p, its roots are u, u^p, u^{p^2}. Their ratios have orders dividing p^2+p+1, whose 3-adic valuation is 1. Raising to 3^r removes this 3-primary part. Lemma 5.2 then gives recurrent divisibility by p.

If the factorization type is $1+2$, the roots lie in $\mathbb{F}_{p^2}$, with one in $\mathbb{F}_p$ and the other two conjugate. Since $v_3(p^2-1) = r+1$, the 3-primary parts of their 3^r-th powers belong to μ_3. The two conjugate roots have the same such part because $p \equiv 1 \pmod 3$. The product of all three parts is 1, by the norm -1, so the remaining part is the same as well. Every ratio consequently has order prime to 3, and Lemma 5.2 again applies.

In either nonsplit case, an arithmetic progression of later traces is divisible by p; these traces eventually exceed p, so the corresponding floors are composite. This contradicts an eventually prime tail. □

The theorem does not exclude completely split prime traces. In particular, it does not show that every candidate has a recurrent prime divisor.

5.3 A prime divisor that never returns

Proposition 5.4 (A completely split nonreturning seed)**.** *Let β be the largest root of*

$$f(X) = X^3 - 211X^2 - 7X + 1, \qquad T_r = \mathrm{Tr}(\beta^{3^r}).$$

It is a totally real cubic Pisot number with conjugates $\beta, -\rho, \sigma$, where

$$0 < \sigma < \rho < 1, \qquad \rho \le \min(\sigma^{17/23}, \beta^{-17/40}).$$

Every T_r equals $\lfloor \beta^{3^r} \rfloor$. Both $T_0 = 211$ and $T_1 = p = 9\,398\,359$ are prime, and f splits completely modulo p, but

$$p \nmid T_r \qquad (r \ge 2).$$

Proof. Sign changes locate roots in $(-1/10, 0), (0, 1), (211, 212)$. The rational-root test excludes ± 1, so f is irreducible. Its root sum and norm give

$$\rho - \sigma = \beta - 211 > 0, \qquad \rho\sigma = \beta^{-1}.$$

Moreover $212^{17} < 10^{40}$, whence $\rho < 1/10 < \beta^{-17/40}$. Combining this with the norm identity gives $\rho^{23} < \sigma^{17}$. For every positive odd integer n, $-\rho^n + \sigma^n \in (-1, 0)$, which proves the floor identity.

Newton's formulas give $T_1 = p$ and $B_1 = -4771$. Trial division to the respective integer square roots certifies the two primality assertions. The three distinct roots modulo p are

$$-1\,584\,786, \qquad -1\,942\,624, \qquad 3\,527\,621.$$

For norm -1, the modular state map is

$$\Phi(t,b) = (t^3 - 3tb - 3,\ b^3 + 3tb + 3) \pmod{p}.$$

Starting at the index-1 state $(0, -4771 \bmod p)$, its next state is

$$S_1 = (9\,398\,356,\ 7\,830\,596).$$

The integer verifier reproduced in Appendix B checks that the orbit from S_1 has exactly 85 712 distinct states, returns to S_1, and has no zero first coordinate. This finite exhaustive certificate proves nondivisibility at every index $r \geq 2$. □

This is a counterexample to universal return of a chosen prime seed, not to the existence of composite terms. Indeed,

$$T_2 = 830\,149\,079\,789\,343\,894\,643 = 7 \cdot 31 \cdot 3\,825\,571\,796\,264\,257\,579.$$

The example also satisfies $a \equiv 1$, $b \equiv -1 \pmod 3$ and $v_3(a+b) = 1$. It shows why the local restrictions above do not by themselves force return of an arbitrarily selected prime trace.

6 Quantitative limits of the transfer to the original constant

6.1 Keeping three prime outputs versus forcing polynomial escape

This subsection analyzes the particular perturbation used in the discussion. It is not an impossibility theorem for all perturbation methods.

Lemma 6.1 (Necessary bounds for a preserved triple)**.** *Work on the hypothetical algebraic Mills tail of Theorem 4.3, and put $a = -\log\rho/\log\beta \in (17/40, 1/2)$. Let B stay in a fixed compact subinterval of $(1,\infty)$. Suppose the clock $3^k - t$, at three consecutive indices $k, k+1, k+2$, has the same outputs p, q, r as the unshifted tail, with $p \to \infty$. Then*

$$t > 0 \implies t = O(p^{-3(1+a)}), \qquad t < 0 \implies |t| = O(p^{-3}).$$

In particular, $|t|p^3 = O(1)$.

Proof. For any two preserved outputs p, q, the exponent identity $(3^{k+1} - t) - 3(3^k - t) = 2t$ gives the exact inequalities

$$\frac{q}{(p+1)^3} < B^{2t} < \frac{q+1}{p^3}. \tag{6.1}$$

Since $q - p^3 \asymp p^{2-a}$, the upper bound implies $t = O(p^{-1-a})$ when $t > 0$. The lower bound, whose logarithm is $-3/p + O(p^{-1-a} + p^{-2})$, implies $|t| = O(p^{-1})$ when $t < 0$. Apply the same estimates to the pair q, r, using $q \sim p^3$ and the same fixed a. □

For comparison, the fixed polynomial (4.2) has a directly verifiable quantitative escape estimate.

Proposition 6.2 (One sufficient escape scale for P_c)**.** *Fix $c = \pm 1$, let $s \to \infty$, $L \to 1$, and put*

$$u = \lfloor s \rfloor, \qquad v = \left\lfloor Ls^3 \right\rfloor, \qquad w = \left\lfloor L^4 s^9 \right\rfloor.$$

If $|L-1|s^3 \to \infty$, then $P_c(u,v,w) \neq 0$ eventually.

Proof. Exact substitution gives

$$P_c(s, Ls^3, L^4s^9) = (L-1)\{s^{12}L(4L-1)(2L+1) + 9cs^9L(L+2) \\ - 27c^2s^6L + 27c^3s^3\}.$$

The leading bracket is asymptotic to $9s^{12}$. Writing the weighted-degree-12 part of P_c as

$$H(x, y, z) = y(x^3 - y)^3 + 9x^3(z - y^3),$$

one checks $H_x(s, s^3, s^9) = H_{xx}(s, s^3, s^9) = 0$. Taylor expansion in the first coordinate, and first-order bounds in the other two, therefore give the uniform floor-error estimate

$$P_c(u, v, w) - P_c(s, Ls^3, L^4s^9) = O(|L-1|s^{11} + s^9).$$

The terms of lower weighted degree contribute only $O(s^9)$. Under the stated hypothesis this error is $o(|L-1|s^{12})$, proving nonvanishing. □

For a perturbed clock, $s = B^{3^k-t}$ and $L = B^{2t}$. When B remains in the stated compact set and $t \to 0$, $|L-1| \asymp |t|$. Thus the sufficient condition just proved requires $|t|p^3 \to \infty$, whereas preservation of the three outputs gives $|t|p^3 = O(1)$. These two particular estimates do not overlap. Pointwise fixed-base avoidance cannot be silently made uniform when the base and the shift vary with the index.

6.2 Intermediate prime conditions remove a skipped-level gain

Lemma 6.3 (The ancestor of an integer root cell)**.** *For integers $q, D \geq 1$, the integer part is constant on $[q^{1/D}, (q+1)^{1/D})$, with value $\left\lfloor q^{1/D} \right\rfloor$.*

Proof. If $j = \left\lfloor q^{1/D} \right\rfloor$, then $j^D \leq q < (j+1)^D$. Since both q and $(j+1)^D$ are integers, $q + 1 \leq (j+1)^D$; the half-open endpoint proves the claim. □

Proposition 6.4 (The terminal search interval)**.** *On a sufficiently late least-Mills tail, fix a depth $d \geq 2$. Among prime terminal labels $q < p_{r+d}$, preservation of the first j descendants leaves the geometric search interval*

$$I_{r,d}^{(j)} = (p_{r+j}^{3^{d-j}}, p_{r+d}), \qquad 0 \leq j < d.$$

For $j < d-1$, requiring the next ancestor to be prime restricts this interval to $I_{r,d}^{(j+1)}$. After all intermediate conditions are imposed, the remaining interval is

$$I_{r,d}^{(d-1)} = (p_{r+d-1}^3, p_{r+d}).$$

This statement does not assert the existence of a prime in any of these intervals.

Proof. By Lemma 6.3, a terminal label determines each earlier integer floor exactly. Below $p_{r+j+1}^{3^{d-j-1}}$, its next ancestor lies between p_{r+j}^3 and $p_{r+j+1} - 1$. Every integer in this range is composite, by the least-prime property, including the initial cube. Above this threshold and below p_{r+d}, the next ancestor is exactly p_{r+j+1}: the upper bound follows from the nesting inequality $p_{r+d} < (p_{r+j+1} + 1)^{3^{d-j-1}}$. Iteration proves the claim. □

The size of the discarded portion can be quantified. With $a = -\log\rho/\log\beta$, $X = \beta^{3^{r+d}}$, and fixed d, j, binomial expansion gives

$$|I_{r,d}^{(j)}| \sim 3^{d-j} X^{\,1-(1+a)/3^{d-j}}.$$

For example, a raw two-level exponent lies in $(5/6, 101/120)$, whereas the final exponent is $\kappa \in (1/2, 21/40)$. More precisely,

$$\frac{|I_{r,d}^{(d-1)}|}{|I_{r,d}^{(0)}|} \sim 3^{1-d}\left(\frac{\rho}{\beta}\right)^{(3^{d-1}-1)3^r} \longrightarrow 0.$$

To verify the asymptotic, write $p_{r+j} = Y - e_{r+j}$, $Y = \beta^{3^{r+j}}$, where $e_{r+j} \sim \rho^{3^{r+j}}$. The leading term of $Y^{3^{d-j}} - e_{r+d} - (Y - e_{r+j})^{3^{d-j}}$ is $3^{d-j}Y^{3^{d-j}-1}e_{r+j}$; the other terms are smaller. Thus the apparent large-interval improvement is removed by the intermediate prime constraints [18].

A collection of many intervals would require an additional quantitative statement about their number and distribution. No such statement sufficient for the present sparse orbit was established in the discussion.

6.3 The analytic input that would finish this route

Let $N(\sigma, T)$ count zeta zeros with real part at least σ and imaginary part in $(0, T]$. The density hypothesis is the estimate

$$N(\sigma, T) \ll_\epsilon T^{2(1-\sigma)+\epsilon} \qquad (1/2 \le \sigma \le 1).$$

Saito proves that DH, and hence RH, implies the transcendence of M [5, Theorem 1.8]. The role of the short-interval input is transparent here: an available prime between x and $x + x^\theta$, for every fixed $\theta > 1/2$ and every sufficiently large x, can be applied with $1/2 < \theta < \kappa$. It then places a prime strictly between p_r^3 and p_{r+1}, contradicting Lemma 4.2.

The OpenAI scope document discussed in the conversation states a zero-free half-plane $\Re s > 7/8$ for the zeta function and the listed L-functions [9]. That scope statement does not supply the density estimate displayed above, or a prime-insertion theorem at every exponent $\theta > 1/2$. No deduction of the required input from that statement is provided here. It is therefore not used as a replacement for DH or RH.

The two unresolved transfers are explicit: a shifted polynomial-avoidance theorem has not been transferred to the actual unshifted minimum, and the necessary prime in its hypothetical algebraic gap has not been forced unconditionally.

7 Reciprocal series and an irrationality transfer criterion

This section supplies the direct connection between the two themes. It covers Mills item M2 and gamma item G6. The conclusions concern reciprocal series built from shifted power floors; an identity expressing Euler's constant by one of these series is a separate, unproved requirement.

7.1 Four clocks and an integer-tail asymptotic

Fix $A > 1$, and consider one of the four exponent sequences

$$C_k = 3^k + \varepsilon \quad \text{or} \quad C_k = 3^k + \varepsilon k, \qquad \varepsilon \in \{-1, 1\}, \quad k \ge 1. \tag{7.1}$$

Put $p_k = \left\lfloor A^{C_k} \right\rfloor$. For the intended applications these integers are primes. In the proofs it is enough that $p_k \geq 1$; they are at least 2 for all sufficiently large k. Define

$$S_1 = \sum_{k\geq 1} \frac{1}{p_k}, \qquad S_2 = \sum_{k\geq 1} \frac{1}{p_k^2}, \qquad P_n = \prod_{k=1}^{n} p_k.$$

Both series converge absolutely. Finite initial terms make no difference to linear independence together with 1.

Lemma 7.1 (Normalized product and tail)**.** *The product*

$$\Omega = \prod_{k\geq 1} \left(\frac{p_k}{A^{C_k}} \right)^2$$

converges to a positive real number. If a, b are fixed real numbers with $a \neq 0$, then, for some $c_0 > 0$,

$$P_n^2 \sum_{k>n} \left(\frac{a}{p_k} + \frac{b}{p_k^2} \right) = a\Omega A^{d_n} + O(\exp(-c_0 3^n)), \tag{7.2}$$

where

$$d_n = 2\sum_{k=1}^{n} C_k - C_{n+1} = \begin{cases} -3 + \varepsilon(2n-1), & C_k = 3^k + \varepsilon, \\ -3 + \varepsilon(n^2-1), & C_k = 3^k + \varepsilon k. \end{cases} \tag{7.3}$$

The implied constants may depend on A, a, b and the chosen clock.

Proof. The relative error in a floor is $p_k/A^{C_k} = 1 + O(A^{-C_k})$. The sum of these errors converges, and no factor is zero. The usual estimate for a convergent infinite product gives

$$P_n^2 = \Omega A^{2\sum_{k\leq n} C_k} (1 + O(A^{-C_{n+1}})).$$

The first tail term satisfies $a/p_{n+1} = aA^{-C_{n+1}}(1 + O(A^{-C_{n+1}}))$. The b-term at that index is $O(A^{-2C_{n+1}})$, and the remaining tail is $O(A^{-C_{n+2}})$. Since $C_{n+2}/C_{n+1} \to 3$, these estimates yield the claimed main term with error $O(A^{d_n - C_{n+1}})$. Formula (7.3) follows by summing the exponents. Because $d_n = O(n^2)$ and C_{n+1} grows as 3^{n+1}, the error is bounded by $O(\exp(-c_0 3^n))$ after decreasing c_0 if necessary. □

7.2 Linear independence of two reciprocal moments

Theorem 7.2 (Two reciprocal series)**.** *For either negative clock $3^k - 1$ or $3^k - k$, and every $A > 1$,*

$$\dim_{\mathbb{Q}} \langle 1, S_1, S_2 \rangle = 3.$$

For either positive clock $3^k + 1$ or $3^k + k$, the same conclusion holds provided $A^2 \notin \mathbb{Q}$. In particular it holds for each of the four shifted prime-generating bases when that base is transcendental.

Proof. Suppose $c + aS_1 + bS_2 = 0$ with integers a, b, c. Then

$$\begin{aligned} I_n &= P_n^2 \sum_{k>n} \left(\frac{a}{p_k} + \frac{b}{p_k^2} \right) \\ &= -cP_n^2 - \sum_{k\leq n} \left(a\frac{P_n^2}{p_k} + b\frac{P_n^2}{p_k^2} \right) \in \mathbb{Z}. \end{aligned}$$

If $a = 0$ and $b \neq 0$, these are nonzero integers, while

$$|I_n| \ll A^{2\sum_{k\le n} C_k - 2C_{n+1}} = A^{d_n - C_{n+1}} \longrightarrow 0.$$

This is impossible. It remains to consider $a \neq 0$. For all sufficiently large k, the sign of $a/p_k + b/p_k^2$ equals the sign of a, so $I_n \neq 0$ eventually. Lemma 7.1 applies.

For the two negative clocks, $d_n \to -\infty$, and again $0 < |I_n| \to 0$ contradicts integrality.

For $C_k = 3^k + 1$, we have $d_n = 2n - 4$. The integer

$$J_n = I_n I_{n+2} - I_{n+1}^2$$

tends to zero: its leading terms cancel, and every remaining term is $O(\exp(O(n) - c_0 3^n))$. Thus $J_n = 0$ for all sufficiently large n. Since $I_n \neq 0$, the rational ratios I_{n+1}/I_n are then constant. Their limit, computed from (7.2), is A^2. This contradicts $A^2 \notin \mathbb{Q}$.

For $C_k = 3^k + k$, we have $d_n = n^2 - 4$. Instead use the integer

$$J_n = I_n I_{n+2}^3 - I_{n+1}^3 I_{n+3}.$$

The identity $d_n + 3d_{n+2} = 3d_{n+1} + d_{n+3}$ cancels the leading terms. The remaining error is $O(\exp(O(n^2) - c_0 3^n))$, which tends to zero. Therefore $J_n = 0$ eventually. The nonzero rational quantities

$$R_n = \frac{I_n I_{n+2}}{I_{n+1}^2}$$

then satisfy $R_{n+1} = R_n$, whereas their limit is $A^{d_n + d_{n+2} - 2d_{n+1}} = A^2$. This is the same contradiction. Consequently $a = b = 0$, and the original relation forces $c = 0$. □

Corollary 7.3. *Under the hypotheses of Theorem 7.2, every nonzero rational linear combination $aS_1 + bS_2$ is irrational.*

Proof. A rational value would give a nontrivial rational relation among $1, S_1, S_2$. □

Remark 7.4 (Scope and the role of primality). The theorem applies to each shifted sequence separately. It does not prove joint independence across the four different sequences, algebraic independence of S_1, S_2, or transcendence of either series. Primality is not needed in the proof: the decisive inputs are the floor approximation and the exponent clock. The positive cases need only $A^2 \notin \mathbb{Q}$, which is weaker than the transcendence hypothesis used in the original discussion. No assertion of historical priority is made for this integer-tail argument.

7.3 A sparse expansion criterion for irrationality

Theorem 7.5 (Bounded integer coefficients on negative clocks)**.** *Let $C_k = 3^k - 1$ or $C_k = 3^k - k$, let $A > 1$, and put $p_k = \left\lfloor A^{C_k} \right\rfloor$. Suppose $r \in \mathbb{Q}$ and $a_k \in \mathbb{Z}$ satisfy*

$$|a_k| \le \sqrt{p_k}, \qquad a_k \neq 0 \text{ for infinitely many } k.$$

Then the absolutely convergent number

$$x = r + \sum_{k\ge 1} \frac{a_k}{p_k}$$

is irrational.

Proof. Suppose $x \in \mathbb{Q}$, and choose an integer $b > 0$ clearing the denominators of both x and r. Then

$$J_n = bP_n \sum_{k>n} \frac{a_k}{p_k} = bP_n(x-r) - b\sum_{k\le n} a_k \frac{P_n}{p_k} \in \mathbb{Z}.$$

Because $C_{k+1}/C_k \to 3$,

$$|J_n| \le bP_n \sum_{k>n} p_k^{-1/2} \ll A^{\sum_{k\le n} C_k - C_{n+1}/2} = A^{d_n/2} \longrightarrow 0.$$

For every sufficiently large j with $a_j \neq 0$, integrality of a_j gives $|a_j/p_j| \ge p_j^{-1}$, while

$$\sum_{k>j} \left|\frac{a_k}{p_k}\right| \le \sum_{k>j} p_k^{-1/2} \ll p_{j+1}^{-1/2} = o(p_j^{-1}).$$

Hence the tail beginning at j is nonzero, so $J_{j-1} \neq 0$ for infinitely many j. This contradicts $J_n \in \mathbb{Z}$ and $J_n \to 0$. □

Corollary 7.6 (The precise missing gamma identity)**.** *If Euler's constant admits an identity*

$$\gamma = r + \sum_{k\ge 1} \frac{a_k}{p_k}$$

satisfying all hypotheses of Theorem 7.5, then γ is irrational.

The theorem proves the implication; the required representation of γ has not been constructed here. A formal expansion with arbitrary real coefficients would not suffice. Neither would an expansion whose integer coefficients violate the stated square-root bound. These are the exact requirements needed to preserve integrality, smallness, and nonvanishing simultaneously.

7.4 What the four transcendence inputs do not imply

Individual transcendence of four constants does not imply their joint algebraic independence: for a transcendental t, the four numbers $t, t+1, t^2, t^3$ are all transcendental and are algebraically dependent. Likewise, the arithmetic property of a limit does not follow from that property of every term: $1 + \pi/n$ is transcendental for every n but converges to 1. These elementary observations explain why an exact transfer such as Corollary 7.6, or the quantified structural statements in the other sections, is necessary.

8 Primitive arithmetic of logarithm-free approximants

This section contains the approximation and denominator results denoted G7–G10 in the result register. It separates three different assertions: an exact positive error formula, a theorem excluding a specified asymptotic range of degrees, and a finite exact computation including degrees not controlled by that asymptotic consequence. None of these assertions proves that γ is rational or irrational.

8.1 The arithmetic target

Proposition 8.1 (A sufficient irrationality criterion)**.** *Suppose that $P_j \in \mathbb{Z}$, $Q_j \in \mathbb{N}$, and*

$$0 < |Q_j\gamma - P_j| \longrightarrow 0.$$

Then $\gamma \notin \mathbb{Q}$.

Proof. If $\gamma = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, then $b(Q_j\gamma - P_j) = aQ_j - bP_j$ is a nonzero integer tending to zero. □

Thus convergence of P_j/Q_j to γ is not by itself the arithmetic target. The error must still tend to zero after multiplication by an integer denominator. Taking the denominator in lowest terms gives the most favorable normalization for that particular rational approximant.

8.2 Binet's measure and its Gaussian approximants

For $s > 0$, put

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad S(s) = \int_0^\infty \frac{d\mu(u)}{s+u}. \tag{8.1}$$

Every moment is finite and rational:

$$\mu_j := \int_0^\infty u^j \, d\mu(u) = \frac{2\Gamma(2j+2)\zeta(2j+2)}{(2\pi)^{2j+2}} = \frac{|B_{2j+2}|}{2j+2} \qquad (j \geq 0). \tag{8.2}$$

Indeed, substitute $u = t^2$, expand $(e^{2\pi t} - 1)^{-1}$ into its positive geometric series, and integrate termwise. Euler's formula for the even zeta values gives the last equality. The first four moments are $1/12, 1/120, 1/252, 1/240$.

The differentiated Binet formula [3, Eq. 5.9.15], together with $\psi(n) = H_n - \gamma - 1/n$, gives

$$H_n - \frac{1}{2n} + S(n^2) = \gamma + \log n \qquad (n \in \mathbb{N}). \tag{8.3}$$

Let $p_m \in \mathbb{Q}[u]$ be the monic orthogonal polynomial of degree m for μ, and set $p_0 = 1$. The moment matrices are positive definite, so these polynomials exist uniquely. All their zeros are simple and belong to $(0, \infty)$; this follows from the usual sign-change argument for orthogonality with a strictly positive measure. Define

$$R_m(s) = \frac{1}{p_m(-s)} \int_0^\infty \frac{p_m(-s) - p_m(u)}{s+u} \, d\mu(u), \qquad R_0 = 0. \tag{8.4}$$

Polynomial division shows that $R_m \in \mathbb{Q}(s)$, with numerator degree at most $m-1$ and denominator degree m. It is the $[m-1/m]$ Gaussian Padé approximant of the formal moment expansion $S(s) \sim \sum_{j\geq 0}(-1)^j \mu_j s^{-j-1}$ at infinity. No convergence of that formal expansion is assumed.

Proposition 8.2 (Positive error and a rational lower certificate)**.** *For $m \geq 0$ and $s > 0$,*

$$E_m(s) := S(s) - R_m(s) = \frac{1}{p_m(-s)^2} \int_0^\infty \frac{p_m(u)^2}{s+u} \, d\mu(u) > 0. \tag{8.5}$$

The function E_m is strictly decreasing on $(0, \infty)$. Moreover,

$$\begin{aligned} 0 < R_{m+1}(s) - R_m(s) &= \frac{h_m}{V_m(s)V_{m+1}(s)} < E_m(s), \\ h_m = \int p_m^2 \, d\mu, &\qquad V_m(s) = (-1)^m p_m(-s) > 0. \end{aligned} \tag{8.6}$$

Proof. Subtract (8.4) from $S(s)$. Orthogonality gives

$$\int p_m(u) \frac{p_m(-s) - p_m(u)}{s+u} \, d\mu(u) = 0,$$

because the quotient is a polynomial in u of degree $m-1$. This proves (8.5). Its numerator integral strictly decreases with s, while $p_m(-s)^2$ increases for $m \geq 1$; for $m = 0$ the numerator alone is strictly decreasing.

For completeness, write the monic three-term recurrence as

$$p_{m+1}(u) = (u - a_m)p_m(u) - b_m p_{m-1}(u), \qquad a_m = \frac{\int u p_m^2 \, d\mu}{h_m}, \quad b_m = \frac{h_m}{h_{m-1}} > 0 \quad (m \geq 1).$$

The denominator polynomials V_m and the corresponding numerators U_m of $R_m = U_m/V_m$ satisfy the same recurrence $X_{m+1} = (s + a_m)X_m - b_m X_{m-1}$, with

$$V_0 = 1, \quad V_1 = s + a_0, \qquad U_0 = 0, \quad U_1 = \mu_0.$$

Consequently $U_{m+1}V_m - U_m V_{m+1} = h_m$ by induction. This gives the middle identity in (8.6); the last inequality follows from $E_m = R_{m+1} - R_m + E_{m+1}$. □

8.3 An exact two-scale logarithm cancellation

Let $N \geq 1$ and $n = 2^N$. Define

$$\begin{aligned} r_{m,N} = (N+1)\left(H_n - \frac{1}{2n} + R_m(n^2)\right) \\ - N\left(H_{2n} - \frac{1}{4n} + R_m(4n^2)\right). \end{aligned} \tag{8.7}$$

This construction, used in [12], makes the logarithms in (8.3) cancel exactly: $(N+1)\log n - N\log(2n) = 0$.

Theorem 8.3 (Logarithm-free positive rational approximants)**.** *For every $m, N \geq 1$, one has $r_{m,N} \in \mathbb{Q}$ and*

$$\begin{aligned} \gamma - r_{m,N} &= (N+1)E_m(n^2) - NE_m(4n^2) \\ &= E_m(n^2) + N\{E_m(n^2) - E_m(4n^2)\} \\ &> E_m(n^2) > R_{m+1}(n^2) - R_m(n^2) > 0. \end{aligned} \tag{8.8}$$

In particular, if $r_{m,N} = P_{m,N}/Q_{m,N}$ is in lowest terms with $Q_{m,N} > 0$, then

$$Q_{m,N}\gamma - P_{m,N} > L_{m,N} := Q_{m,N}\{R_{m+1}(n^2) - R_m(n^2)\} > 0, \tag{8.9}$$

and $L_{m,N}$ is a rational number that can be compared exactly with 1.

Proof. Substitute (8.3) into (8.7), then apply Proposition 8.2. Rationality follows from the rationality of all the moments. □

The approximants do converge to γ as $N \to \infty$, even if $m = m(N)$ varies arbitrarily in $\mathbb{N}$, since

$$0 < \gamma - r_{m,N} < (N+1)S(n^2) \leq \frac{(N+1)\mu_0}{n^2}.$$

The five requirements in the research program therefore have the following exact status.

Requirement	Status for (8.7)
Exact removal of logarithms	Established.
Strictly nonzero error	Established by positivity.
Error tends to zero after multiplication by the reduced denominator	Not established; see the results below.
Infinitely many approximants	Established.
No unproved auxiliary hypothesis	Established for the construction.

8.4 A denominator barrier for slowly increasing degrees

Write $\vartheta(x) = \sum_{\ell \le x,\ \ell \text{ prime}} \log \ell$. The following statement is the slow-degree result in [12]; we give its denominator and error arguments explicitly.

Theorem 8.4 (Slow-degree Stieltjes barrier)**.** *There is an absolute constant $C > 0$ such that, for every $m, N \ge 1$ and $n = 2^N$,*

$$\begin{aligned}\log\{Q_{m,N}(\gamma - r_{m,N})\} \ge \vartheta(2n) - \vartheta(n) \\ - C\{m^2 \log(m+1) + (m+1)\log n\}.\end{aligned} \tag{8.10}$$

Consequently, as $N \to \infty$,

$$m^2 \log(m+1) + (m+1)\log n = o(n) \quad \Longrightarrow \quad Q_{m,N}(\gamma - r_{m,N}) \longrightarrow +\infty. \tag{$*$}$$

In particular, $(*)$ holds if $m = o(\sqrt{n/\log n})$.

Proof. We first control the rational coefficients independently of n. The von Staudt–Clausen denominator theorem shows that a common denominator $\mathfrak{d}_m$ of $\mu_0, \dots, \mu_{2m}$ can be chosen to divide

$$\operatorname{lcm}(1, \dots, 4m+2) \prod_{\ell \le 4m+3,\ \ell \text{ prime}} \ell.$$

Thus $\log \mathfrak{d}_m = O(m)$. The factorial expression in (8.2) implies that the integerized moments have logarithmic size $O(m \log(m+1))$. Apply Cramer's rule to the m equations $\int u^j p_m(u)\, d\mu(u) = 0$, $0 \le j < m$. After multiplication of the moment matrix by $\mathfrak{d}_m$, its determinant is a nonzero integer, and every determinant involved is bounded by $m!$ times an mth power of the largest entry. It follows that the rational coefficients of p_m, and their numerators and denominators, have logarithmic height $O(m^2 \log(m+1))$.

In particular, there are integer polynomials $\widetilde{U}_m, \widetilde{V}_m$ of coefficient logarithmic height $O(m^2 \log(m+1))$, with $\deg \widetilde{V}_m = m$, such that

$$R_m(s) = \frac{\widetilde{U}_m(s)}{\mathfrak{d}_m \widetilde{V}_m(s)}.$$

The polynomial $\widetilde{V}_m$ is an integer multiple of $p_m(-s)$, so it has no positive real zero. Its evaluation satisfies

$$\log |\widetilde{V}_m(t)| \le C_1 m^2 \log(m+1) + m \log(1+t) \qquad (t > 0).$$

Now let ℓ be a prime with $n < \ell \le 2n$. Since n is a power of 2, the rational terms $1/(2n)$ and $1/(4n)$ are integral at ℓ. The harmonic term H_n is also integral there, whereas H_{2n} has exactly one summand with denominator divisible by ℓ, namely $1/\ell$. Moreover, $\ell > N$. Unless

$$\ell \mid \mathfrak{d}_m \widetilde{V}_m(n^2) \widetilde{V}_m(4n^2),$$

both Padé evaluations are integral at ℓ, and (8.7) has ℓ-adic valuation -1, coming from $-N/\ell$. Every such nonexceptional prime therefore divides its actual reduced denominator $Q_{m,N}$. The total logarithm of the exceptional primes is at most the logarithm of the displayed nonzero integer. Hence

$$\log Q_{m,N} \ge \vartheta(2n) - \vartheta(n) - C_2\{m^2 \log(m+1) + m \log n\}. \tag{8.11}$$

It remains to give a lower bound on the positive error. The density of μ on $[1,2]$ is at least $c_\mu = (e^{2\pi\sqrt{2}} - 1)^{-1} > 0$. Among monic polynomials of degree m on an interval of length 1, the shifted monic Legendre polynomial minimizes the squared L^2 norm. Consequently

$$\int_1^2 p_m(u)^2\, du \ge \frac{1}{(2m+1)\binom{2m}{m}^2} \ge \frac{1}{(2m+1)16^m}.$$

Formula (8.5) now yields

$$E_m(n^2) \geq \frac{c_\mu}{(n^2+2)(2m+1)16^m\, p_m(-n^2)^2}.$$

The coefficient height bound already proved implies

$$\log E_m(n^2) \geq -C_3\{m^2 \log(m+1) + (m+1)\log n\}. \tag{8.12}$$

Combine (8.11), (8.12), and (8.8). Finally the prime number theorem gives $\vartheta(2n) - \vartheta(n) = n + o(n)$, proving $(*)$. □

This is an asymptotic exclusion for the stated degree range and this particular two-scale family. It makes no assertion that every high degree, different linear combination, or different approximation method must fail.

8.5 Exact computation beyond the slow-degree range

To inspect degrees not covered by the preceding asymptotic consequence, the calculation used

$$n \in \{2, 4, 8, 16, 32, 64\}, \qquad 1 \leq m \leq 64.$$

The monic orthogonal recurrence was evaluated with exact rational arithmetic. For each of these 384 pairs, (8.7) was reduced to lowest terms before computing $L_{m,N}$ in (8.9).

Proposition 8.5 (Finite exact certificate)**.** *For all* 384 *pairs in the displayed range,*

$$L_{m,N} > 1, \qquad \textit{and therefore} \qquad Q_{m,N}\gamma - P_{m,N} > 1.$$

Computational proof. The standard-library verifier included in the computational appendix constructs the Bernoulli numbers as rational numbers, computes the monic orthogonal recurrence, and compares the numerator and denominator of each positive rational $L_{m,N}$. The test is an integer comparison; no numerical value of γ is an input. The archived output contains exactly 384 successful comparisons. The small cases $m = 1, 2, 3$, $n = 2, 4$ were additionally checked by solving the moment orthogonality equations directly, independently of that recurrence. □

For an explicit certificate visible without running a large calculation, the first recurrence coefficients are

$$a_0 = \frac{1}{10}, \qquad a_1 = \frac{871}{790}, \qquad b_1 = \frac{79}{2100}.$$

Thus

$$R_1(s) = \frac{5}{6(10s+1)}, \qquad R_2(s) = \frac{(s+871/790)/12}{(s+1/10)(s+871/790) - 79/2100}.$$

At $n = 4$, $N = 2$, $m = 1$ these give

$$r_{1,2} = \frac{3574141}{6192060}, \qquad L_{1,2} = \frac{4000481}{913489} > 4, \tag{8.13}$$

and hence $6192060\gamma - 3574141 > 4000481/913489 > 4$.

n	m	Digits of the reduced $Q_{m,N}$	$\log_{10} L_{m,N}$ (rounded)
4	1	7	0.641409
4	3	27	15.837548
8	8	223	197.775136
16	16	1043	993.168330
32	32	5121	5022.804049
64	64	24705	24508.197207

Only the logarithms in the table are rounded. For example, at $(n, m) = (4, 3)$ the primes 5 and 7 from the harmonic denominator both disappear from the final reduced denominator. The exact test still fails to produce a small linear form. Thus the computation does account for actual cancellation. Its finite range cannot decide the behavior of all (m, N) or of combinations of different approximants.

8.6 The exact lattice of one rank-three CMF trajectory

The family of Apéry-like recurrences in [8] supplies a different setting in which denominator normalization matters. The following calculation concerns the specific trajectory $(0, 0; 1, 1, 1)$ and the projection used in [15]. It is a theorem about its complete image at each depth, not a claim that every depth-dependent input inherits a fixed-input error bound.

Define the integer matrices

$$M_n = \begin{pmatrix} 1 & n^3 & 3(n+1)(n-1)(n^2+n+1) \\ 2 & 1-3n^2 & -(n+1)(8n^2+n+5) \\ 1 & 3n+2 & 6n(n+1) \end{pmatrix}$$

and

$$N_n = \begin{pmatrix} 6n^2+13n+10 & -8n-5 & 3 \\ 3n^3+3n^2-3n-6 & -3n^2+3n+3 & n-2 \\ n^4-2n^3-6n^2-3n+3 & 6n^2+2n-1 & 1-2n \end{pmatrix}.$$

Direct multiplication gives

$$\det M_n = \det N_n = (n+1)^6, \qquad N_n = (n+1)^4 M_n^{-T}. \tag{8.14}$$

Let $P_n = N_0 \cdots N_{n-1}$, $U_n = M_0 \cdots M_{n-1}$, with both empty products equal to I_3, and put

$$R = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \qquad RP_n = \begin{pmatrix} \mathbf{p}_n \\ \mathbf{q}_n \end{pmatrix}, \qquad \Lambda_n = RP_n\mathbb{Z}^3.$$

Theorem 8.6 (Exact image and global content)**.** *With $H_0 = 0$, one has*

$$\Lambda_n = \{(n!a + 3n!H_n b,\ n!b)^T : a, b \in \mathbb{Z}\}. \tag{8.15}$$

If $D_n = \operatorname{den}(3H_n)$ is the reduced positive denominator and $C_n = 3D_nH_n$, then the gcd of all entries of RP_n is $n!/D_n$. Dividing the entire matrix by that common gcd gives the image lattice

$$\{(D_n a + C_n b,\ D_n b)^T : a, b \in \mathbb{Z}\}, \qquad [\mathbb{Z}^2 : \textit{this lattice}] = D_n^2. \tag{8.16}$$

Proof. We give the two arithmetic components of the proof. First, (8.14) implies $P_n = (n!)^4 U_n^{-T}$ and $\det P_n = (n!)^6$. The cross product of the two rows of R is $(1, -1, 0)$, so

$$\operatorname{cross}(\mathbf{p}_n, \mathbf{q}_n) = (n!)^2 \mathbf{b}_n, \qquad \mathbf{b}_n = (1, -1, 0)U_n \in \mathbb{Z}^3. \tag{8.17}$$

The vector $\mathbf{b}_n$ is primitive. To see this, put $\theta = z\,d/dz$, $\mathscr{D} = \theta - z$, and

$$J_n(z) = \left[\prod_{j=0}^{n-1}(j - \mathscr{D})\right]^3 1.$$

The gauge identity $e^z \theta e^{-z} = \mathscr{D}$ gives

$$e^{-z} J_n(z) = z^n \sum_{k \geq 0} \frac{(-1)^k (n+k)!^2}{k!^3} z^k.$$

This function satisfies $((\theta-n)^3+z(\theta+1)^2)f=0$ and is carried to the next one by $z(\theta+1)^2$. Computing the first three jets at $z=1$ gives

$$\mathbf{b}_n=(J_n(1),\mathscr{D}J_n(1),\mathscr{D}^2J_n(1)).$$

Over $\mathbb{F}_\ell[z]$ one has $\mathscr{D}^\ell-\mathscr{D}=-z^\ell$. For example, the left side commutes with multiplication by z, and its value on 1 is $-z^\ell$ by the Stirling-number formula for $\mathscr{D}^\ell 1$ and Fermat's identity. It follows that $\prod_{j=0}^{\ell-1}(j-\mathscr{D})=z^\ell$. Writing $n=a\ell+r$ with $0\le r<\ell$ therefore gives $\mathbf{b}_n\equiv\mathbf{b}_r \pmod{\ell}$. Every M_j, $0\le j<r$, is invertible modulo ℓ, so $\mathbf{b}_r\not\equiv 0$. No prime divides all entries of $\mathbf{b}_n$. By (8.17), the gcd of the maximal minors of RP_n is $(n!)^2$, which is the covolume of Λ_n.

Second, the harmonic shear of the two rows is integral. Here are finite formulas making this assertion explicit. Put $D=u\,d/du$ and

$$Q_n(u)=\sum_{k=0}^{n}\binom{n}{k}^3 k!\,u^k,\qquad A_n(u)=\sum_{k=0}^{n}\binom{n}{k}^3 k!\{3(H_{n-k}-H_n)-2H_k\}u^k.$$

Both polynomials have integer coefficients: $k!H_k$ is integral, and $\binom{n}{k}^3k!=\binom{n}{k}^2n!/(n-k)!$ is divisible by each integer between $n-k+1$ and n. Set $a_n=1-3n$, $d_n=1+3n^2$, and define integer rows

$$\begin{aligned}\mathcal{R}_{Q,n}&=\Big((D^2+a_nD+d_n)Q_n,\ (D+a_n)Q_n,\ Q_n\Big)_{u=1},\\ \mathcal{R}_{A,n}&=\big((D^2+a_nD+d_n)A_n+2DQ_n+a_nQ_n,\\ &\qquad (D+a_n)A_n+Q_n,\ A_n\big)_{u=1}.\end{aligned}\tag{8.18}$$

These rows satisfy

$$\begin{aligned}\mathcal{R}_{Q,n+1}&=\frac{\mathcal{R}_{Q,n}N_n}{n+1},\\ \mathcal{R}_{A,n+1}&=\frac{\mathcal{R}_{A,n}N_n}{n+1}-\frac{3\mathcal{R}_{Q,n+1}}{n+1}.\end{aligned}\tag{8.19}$$

One verification is to use the adjoint solutions $w_Q=Q_n(1/z)$ and $w_A=A_n(1/z)-\log z\,Q_n(1/z)$ of $z\theta^2w-(\theta+n)^3w=0$. For f solving $((\theta-n)^3+z(\theta+1)^2)f=0$, the expression

$$(\theta^2w-(z-3n)\theta w+(z+3n^2)w)f+(-\theta w+(z-3n)w)\theta f+w\theta^2 f$$

has derivative zero. If $f=z^n(\alpha(\log z)^2/2+\beta\log z+\chi+O(z\log^2 z))$, its values for w_Q,w_A are respectively $n!\alpha$ and $n!(\beta-5H_n\alpha)$. The map $f\mapsto z(\theta+1)^2f$ changes (α,β) to $((n+1)^2\alpha,(n+1)^2\beta+2(n+1)\alpha)$. Evaluating the conserved expression at $z=1$ proves (8.19). The displayed adjoint solutions themselves follow by differentiating at $\varepsilon=0$ the finite sum $z^{-\varepsilon}\sum_{k=0}^{n}c_k(\varepsilon)z^{-k}$, where $c_0=1$ and $c_{k+1}/c_k=(n-\varepsilon-k)^3/(\varepsilon+k+1)^2$; its residual under $(\theta+n)^3-z\theta^2$ is $-\varepsilon^2z^{1-\varepsilon}-\varepsilon^3c_n(\varepsilon)z^{-n-\varepsilon}$.

The initial rows are $\mathcal{R}_{Q,0}=(1,1,1)$ and $\mathcal{R}_{A,0}=(1,1,0)$. Induction in (8.19) gives

$$\mathbf{q}_n=n!\mathcal{R}_{Q,n},\qquad \mathbf{p}_n-3H_n\mathbf{q}_n=n!\mathcal{R}_{A,n}.$$

Hence Λ_n is contained in the right side of (8.15). Both lattices have covolume $(n!)^2$, so they are equal. Finally $\gcd(n!,3n!H_n)=n!/D_n$ proves the content and normalization assertions. □

Remark 8.7 (Global normalization versus an individual reduced pair). The division in Theorem 8.6 removes the gcd shared by *all* entries of the projected matrix. A selected pair (D_na+C_nb,D_nb) can have an additional gcd. For example, if $\gcd(a,b)=1$ and this additional gcd is g, then $g\mid D_n^2$: multiplication by the adjugate of $\left(\begin{smallmatrix}D_n & C_n\\ 0 & D_n\end{smallmatrix}\right)$ shows that g divides both D_n^2a and D_n^2b. It would be incorrect to call every globally normalized output already primitive. Any method using additional per-output cancellation must analyze that cancellation together with its error.

At each fixed n, the nonzero-denominator ratios in (8.16) are exactly $3H_n + a/b$, hence run through all of $\mathbb{Q}$. Unrestricted freedom to change the input is therefore not, on its own, an arithmetic approximation theorem. The globally normalized linear form is

$$\ell_n(a,b) = D_n a + C_n b - \gamma D_n b = D_n\{a - b(\gamma - 3H_n)\}. \tag{8.20}$$

Dirichlet's theorem provides $a, b \in \mathbb{Z}$, $1 \le b \le B$, with $|\ell_n(a,b)| \le D_n/B$. This upper bound tends to zero if $B/D_n \to \infty$. On the other hand, under a reductio assumption $\gamma = r/s$, $\ell_n(a,b) = 0$ would imply $D_n \mid sb$, since $\gcd(C_n, D_n) = 1$. Thus $0 < |b| < D_n/s$ is a sufficient certificate of nonvanishing. Using $B < D_n/s$ as a uniform certificate for the Dirichlet choices is incompatible with making the bound D_n/B tend to zero.

This last observation concerns these two sufficient estimates. It does not rule out stronger approximation estimates, separately proved nonvanishing, useful per-output gcd cancellation, or carefully restricted adaptive inputs. Under $\gamma = r/s$, every nonzero integer-coefficient form is at least $1/s$ in absolute value, while the exact lattice also contains nonzero pairs with form zero. An argument that produces a small form but permits that form to vanish has not supplied Proposition 8.1.

9 The remaining transfer requirements

The preceding sections provide several complete statements about objects surrounding γ and M. Their implications for the two original targets require additional inputs that must not be supplied by analogy. This section records those inputs explicitly.

9.1 Euler's constant

The field-theoretic radical criterion would prove transcendence of γ if one could establish $\delta^{1/n} \in \mathcal{K}$ for some $n \ge 2$. No such membership is proved or predicted here. The relative-closure theorem controls elements already known to lie in $\mathcal{K}$; it does not force an arbitrary analytic constant or one of its roots to belong to $\mathcal{K}$. Likewise, $\delta = e(\eta - \gamma)$ shows that algebraic γ would put δ in the E-value ring, but that membership alone does not violate any of the proved closure statements.

The reciprocal-series route requires the actual identity in Corollary 7.6, with integer coefficients, the stated size bound, and infinitely many nonzero coefficients. The irrationality of the permitted class of series does not identify γ as a member.

For rational approximants the complete success criterion remains

$$P_j \in \mathbb{Z}, \quad Q_j \in \mathbb{N}, \quad 0 < |Q_j\gamma - P_j| \longrightarrow 0.$$

The Padé family supplies logarithm cancellation and nonvanishing for infinitely many indices. The slow-degree theorem rules out the displayed asymptotic degree range, and the 384-case certificate checks a finite collection of larger degrees. Neither result decides every remaining degree schedule or every linear combination of approximants. In the CMF discussion, additional cancellation in a selected output pair has to be analyzed together with its error and its nonvanishing; global matrix normalization is not that analysis.

9.2 The least classical Mills constant

Under hypothetical algebraicity, the least-prime property and the cubic trace formulas force special prime-free intervals immediately above prime cubes. An unconditional prime in one such interval, far enough along the hypothetical tower, would contradict minimality. The given

gap divisibility, exponent, three-adic, and splitting restrictions have not been shown mutually inconsistent.

The shifted clocks supply fixed-polynomial escape, but applying it to a classical algebraic candidate requires preserving the appropriate prime labels at a scale where that escape estimate is effective. The estimates audited here do not furnish such a scale. Taking limits of shifted constants or discarding intermediate prime constraints does not repair that missing step.

9.3 Reusable completed statements

The relative-closure and intersection results, the explicit compositum counterexample, the reciprocal-series linear independence theorem, the bounded-coefficient irrationality criterion, and the exact cubic trace identities stand on their own stated hypotheses. The accompanying certificates establish the specified finite claims with integer or rational arithmetic. These are the completed results of this compilation. The two original arithmetic questions remain outside those conclusions.

A Reproducing the Padé arithmetic certificate

The following Python program uses only the standard library. Save the listing as `pade_certificate.py` and run `python3 pade_certificate.py 64` with Python 3.11 or later. It recomputes all Bernoulli moments and all rational approximants; it does not read precomputed numerical values of γ or a separate input dataset. Every inequality is tested on exact integers through `fractions.Fraction`. The floating-point logarithms are only display values. An assertion checks that every tested lower bound is greater than one. The JSON output is generated by the program and is not required to compile or read this manuscript.

The complete 384-case program was rerun for this edition and agreed entry-for-entry with the preceding computation. Its summary is

$$\text{cases} = 384, \qquad \text{all } L_{m,N} > 1, \qquad \min \log_{10} L_{m,N} = 0.641409 \text{ (rounded)}.$$

The unique source file contains the complete listing, not a reference to an external script.

```
from fractions import Fraction as F
from math import comb, gcd, log10, isqrt
import json
import sys
import time
sys.set_int_max_str_digits(0)
max_m = int(sys.argv[1]) if len(sys.argv)>1 else 64
scales = [2,4,8,16,32,64]
def bernoullis(n):
    out = [F(1)]
    for j in range(1,n+1):
        out.append(-sum((F(comb(j+1,k))*out[k] for k in range(j)),F(0))/F(j+1))
    return out
def primes(a,b):
    return [p for p in range(a,b+1) if p>=2 and all(p%d for d in range(2,isqrt(p)+1))]
def harmonic(n):
    return sum((F(1,k) for k in range(1,n+1)),F(0))
def logf(q):
    return log10(q.numerator)-log10(q.denominator)
started=time.monotonic()
B=bernoullis(4*max_m+4)
mu=[abs(B[2*j+2])/F(2*j+2) for j in range(2*max_m+2)]
p_prev=[F(0)]
p=[F(1)]
h_prev=None
```

```
aa=[]
bb=[F(0)]
R={n:[] for n in scales}
prev={s:(F(0),F(1),F(1),F(0)) for n in scales for s in (n*n,4*n*n)}
history={s:[] for s in prev}
output=[]
for m in range(max_m+1):
    h=sum((p[i]*mu[i+m] for i in range(m+1)),F(0))
    a=sum((p[i]*mu[i+m+1] for i in range(m+1)),F(0))/h+(p[m-1] if m else 0)
    assert h>0 and a>0
    if m:
        b=h/h_prev
        assert b>0
        bb.append(b)
    else:
        b=F(0)
    aa.append(a)
    for s,(N1,N0,D1,D0) in list(prev.items()):
        # m=0 constructs R_1; m>=1 advances the usual Jacobi convergents.
        if m==0:
            N2,D2=mu[0],F(s)+a
        else:
            N2=(F(s)+a)*N1-b*N0
            D2=(F(s)+a)*D1-b*D0
        r=N2/D2
        if history[s]: assert r>history[s][-1]
        history[s].append(r)
        prev[s]=(N2,N1,D2,D1)
    # After R_{m+1}, compare the previous R_m at each scale.
    if m>=1:
        for n in scales:
            N=n.bit_length()-1
            rm=(N+1)*(harmonic(n)-F(1,2*n)+history[n*n][m-1])-N*(harmonic(2*n)-F(1,4*n)+history[4*
    n*n][m-1])
            Q=rm.denominator
            diff=history[n*n][m]-history[n*n][m-1]
            exact_lower=Q*diff
            interval_primes=primes(n+1,2*n)
            removed=[p for p in interval_primes if Q%p]
            row={'n':n,'m':m,'Q_digits':len(str(Q)),
                 'log10_certified_lower':round(logf(exact_lower),6),
                 'certified_lower_gt_1':exact_lower>1,
                 'fresh_primes':interval_primes,'removed_fresh_primes':removed}
            output.append(row)
            if m in {max(1,n//4),max(1,n//2),n,2*n,4*n}:
                print(json.dumps(row),flush=True)
    p_new=[F(0)]*(m+2)
    for i,co in enumerate(p):
        p_new[i+1]+=co
        p_new[i]-=a*co
    if m:
        for i,co in enumerate(p_prev): p_new[i]-=b*co
    p_prev,p=p,p_new
    h_prev=h
assert len(output) == max_m * len(scales)
assert all(x['certified_lower_gt_1'] for x in output)
print(json.dumps({'computed_max_m':max_m,'seconds':round(time.monotonic()-started,3),
                  'cases':len(output),'all_certified_lower_gt_1':all(x['certified_lower_gt_1'] for
     x in output),
                  'smallest_log10_lower':min(x['log10_certified_lower'] for x in output)}),flush=
    True)
with open('pade_certificate_results.json','w') as file:
    json.dump(output,file,indent=2)
```

B Reproducing the prime-seed nonreturn certificate

Save the following standard-library program as `mills_nonreturn_verify.py` and run it with Python 3.11 or later. All root brackets, primality checks, congruences, and orbit statements are verified exactly. The loop checks every one of the 85 712 distinct states, not a sample of the cycle. The certificate establishes nonreturn modulo the specified prime; it does not assert that all subsequent traces are prime. In fact, the last assertions give the explicit factorization of T_2.

```
"""Exact certificate for the nonreturning cubic prime seed.
Python 3 standard library only.  No floating-point arithmetic and
no probabilistic primality tests are used.
"""
from fractions import Fraction
from math import isqrt
def f(x):
    return x**3 - 211*x**2 - 7*x + 1
def is_prime(n):
    if n < 2:
        return False
    if n % 2 == 0:
        return n == 2
    return all(n % d for d in range(3, isqrt(n) + 1, 2))
def step(t, b, modulus=None):
    out = (t**3 - 3*t*b - 3, b**3 + 3*t*b + 3)
    if modulus is None:
        return out
    return tuple(x % modulus for x in out)
assert f(Fraction(-1, 10)) < 0 < f(0)
assert f(0) > 0 > f(1)
assert f(211) < 0 < f(212)
assert f(1) != 0 and f(-1) != 0
assert 212**17 < 10**40
t0, b0 = 211, -7
p, b1 = step(t0, b0)
assert (p, b1) == (9398359, -4771)
assert is_prime(t0) and is_prime(p)
roots = (-1584786, -1942624, 3527621)
assert len({x % p for x in roots}) == 3
assert all(f(x) % p == 0 for x in roots)
start = step(0, b1 % p, p)
assert start == (9398356, 7830596)
state = start
seen = set()
for _ in range(85712):
    assert state not in seen
    seen.add(state)
    assert state[0] != 0
    state = step(*state, p)
assert state == start
assert len(seen) == 85712
t2, _ = step(p, b1)
assert t2 == 830149079789343894643
assert t2 == 7 * 31 * 3825571796264257579
assert (211 - 1) % 3 == 0
assert (-7 + 1) % 3 == 0
assert (211 - 7) % 3 == 0 and (211 - 7) % 9 != 0
print("PASS: exact root brackets, primality, complete splitting,")
print("85712 distinct nonzero-trace cycle states, and T_2 factorization.")
```

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